CMPE 297-01 Computer Vision

Introduction to Cameras

Fall 2026, DMH 348

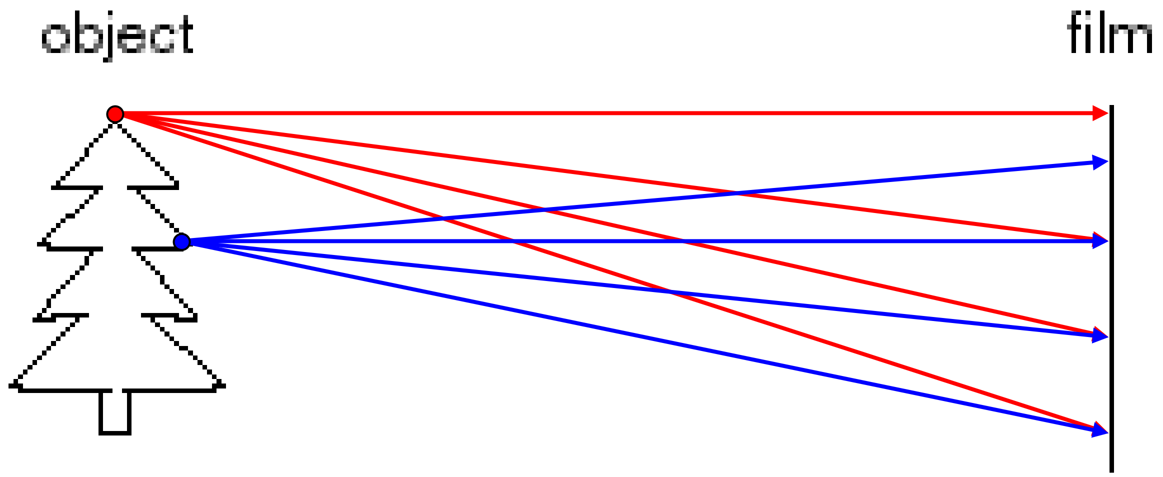

KODAK
DRESTOP SYNCHRO

Lecture Outline

- Pinhole projection
- Lenses
- Field of view, focal length
- Marginal distortion
- Lens distortions

Pinhole Projection

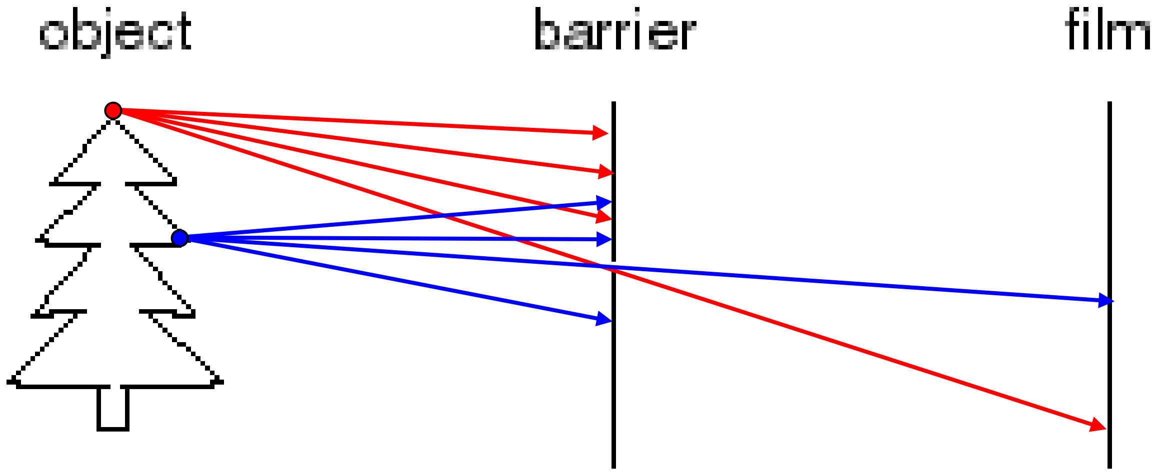
Let’s Design a Camera



Idea 1: put a piece of film in front of an object

Do we get a reasonable image?

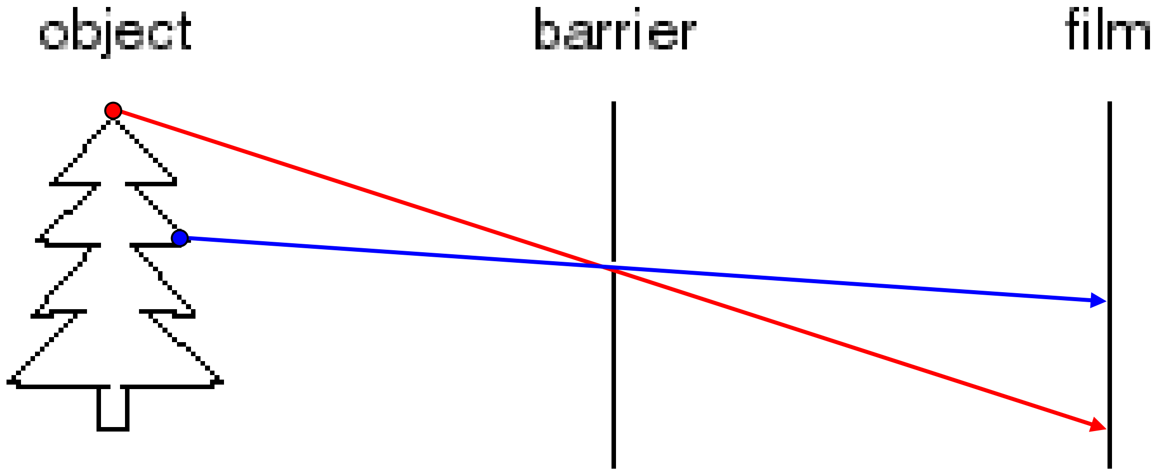
Pinhole Camera



Add a barrier to block off most of the rays

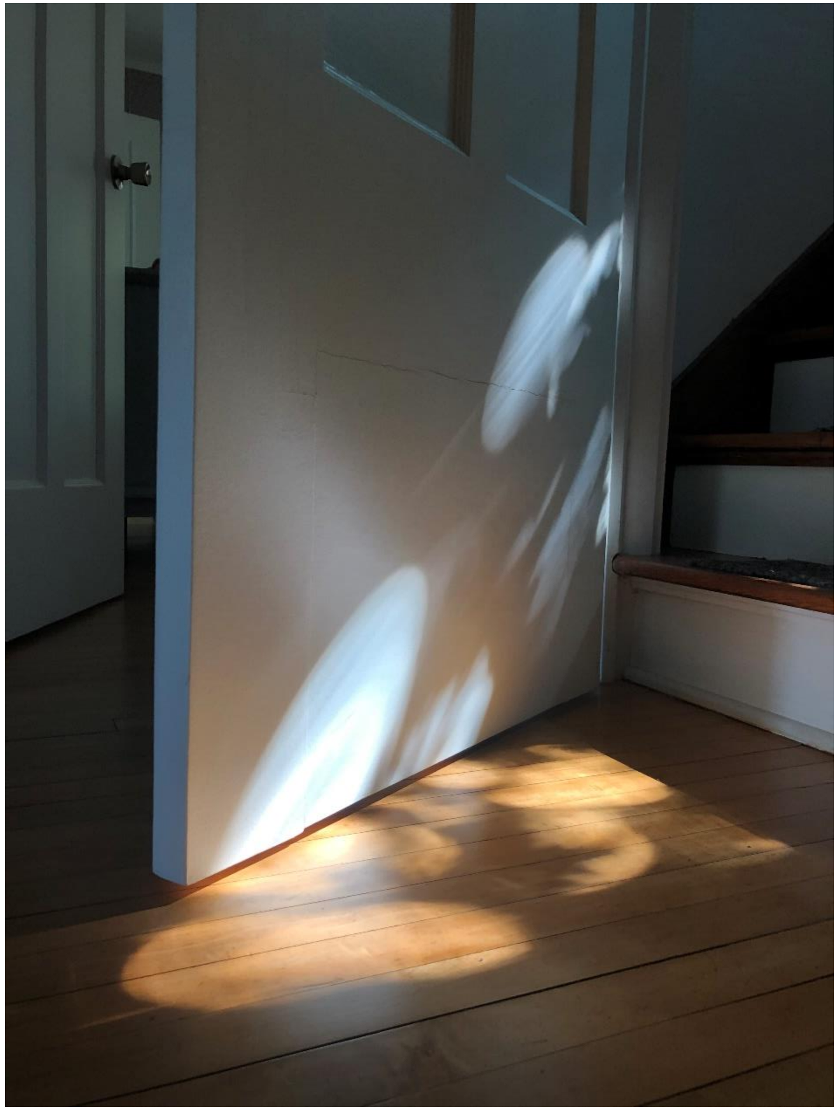
Slide by Steve Seitz

Pinhole Camera

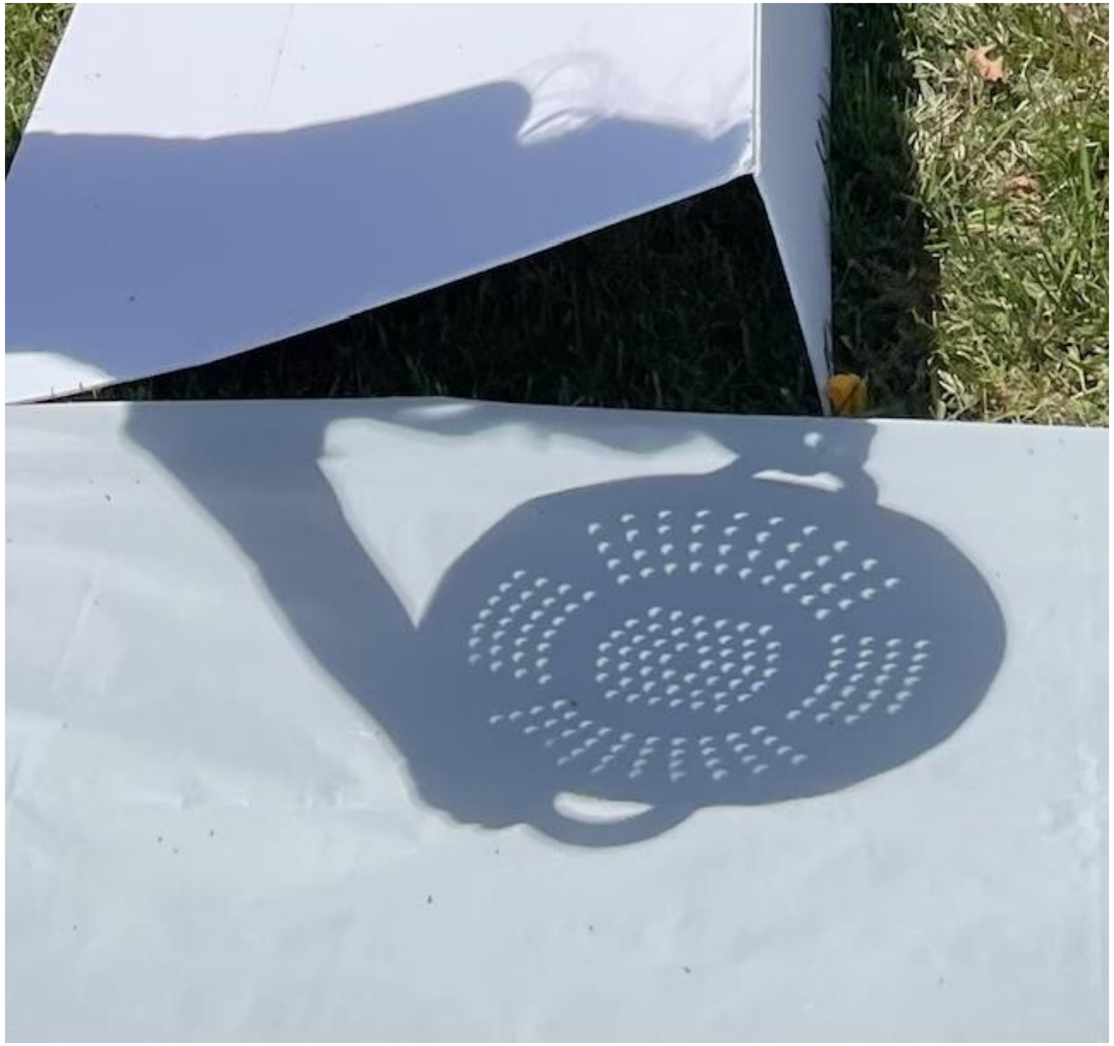


- Captures **pencil of rays** – all rays through a single point: **aperture, center of projection, optical center, focal point, camera center**
- The image is formed on the **image plane**

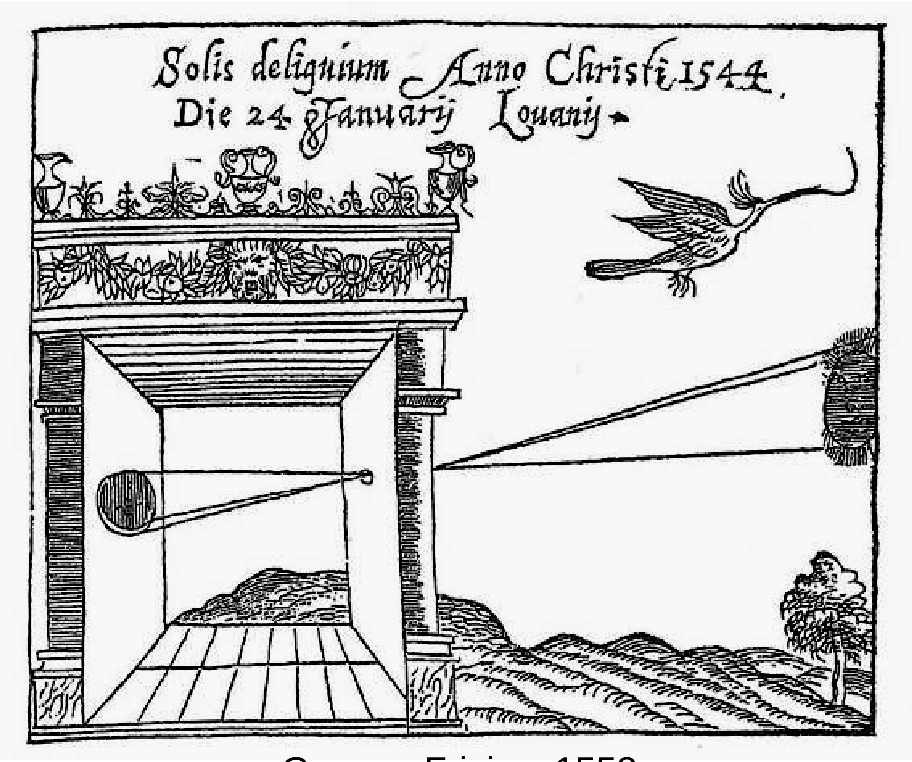
Pinhole Cameras are Everywhere



Pinhole Cameras are Everywhere



Camera Obscura

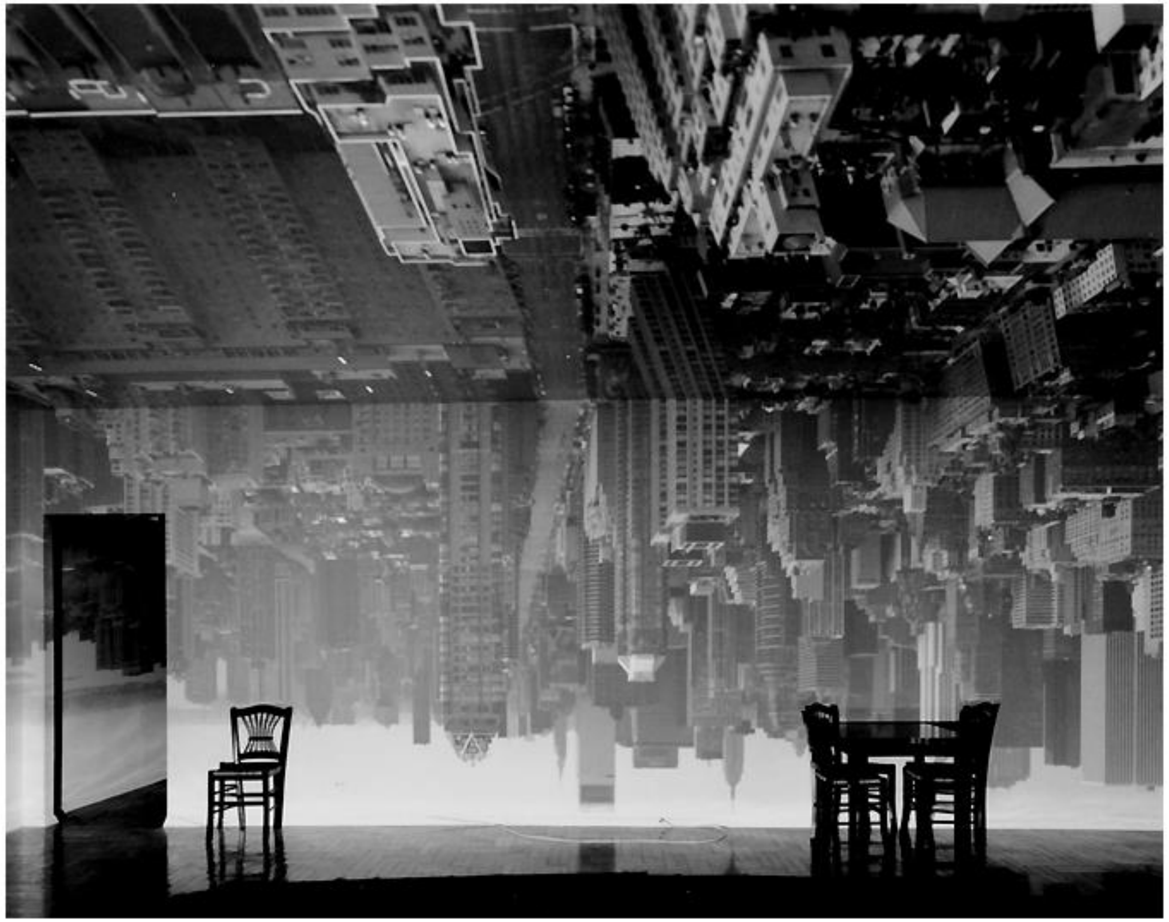


Gemma Frisius, 1558

- Basic principle known to Mozi (470-390 BCE), Aristotle (384-322 BCE)
- Drawing aid for artists: described by Leonardo da Vinci (1452-1519)

Source: A. Efros

Turning a Room Into a Camera Obscura



Abelardo Morell, Camera Obscura Image of Manhattan View Looking South in Large Room, 1996

After scouting rooms and reserving one for at least a day, Morell masks the windows except for the aperture. He controls three elements: the size of the hole, with a smaller one yielding a sharper but dimmer image; the length of the exposure, usually eight hours; and the distance from the hole to the surface on which the outside image falls and which he will photograph. He used 4 x 5 and 8 x 10 view cameras and lenses ranging from 75 to 150 mm.

After he's done inside, it gets harder. "I leave the room and I am constantly checking the weather, I'm hoping the maid reads my note not to come in, I'm worrying that the sun will hit the plastic masking and it will fall down, or that I didn't trigger the lens."

From *Grand Images Through a Tiny Opening*, **Photo District News,** February 2005

https://www.abelardomorell.net/camera-obscura

Turning a Room Into a Camera Obscura

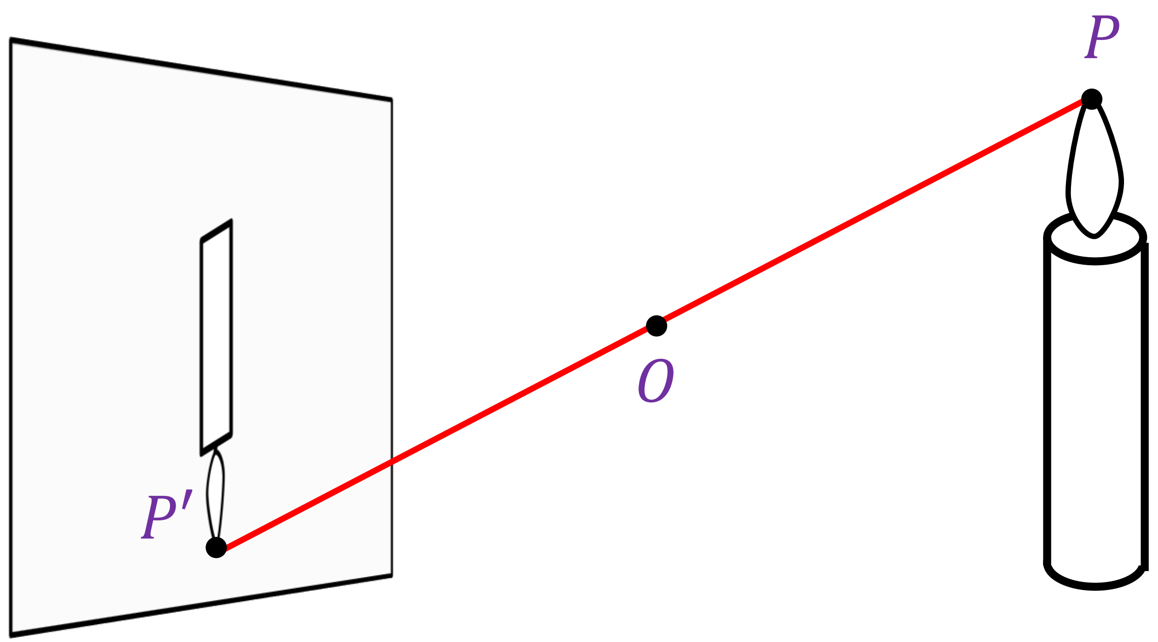
My hotel room,
contrast enhanced.

The view from my window

Accidental pinholes produce images that are unnoticed or misinterpreted as shadows

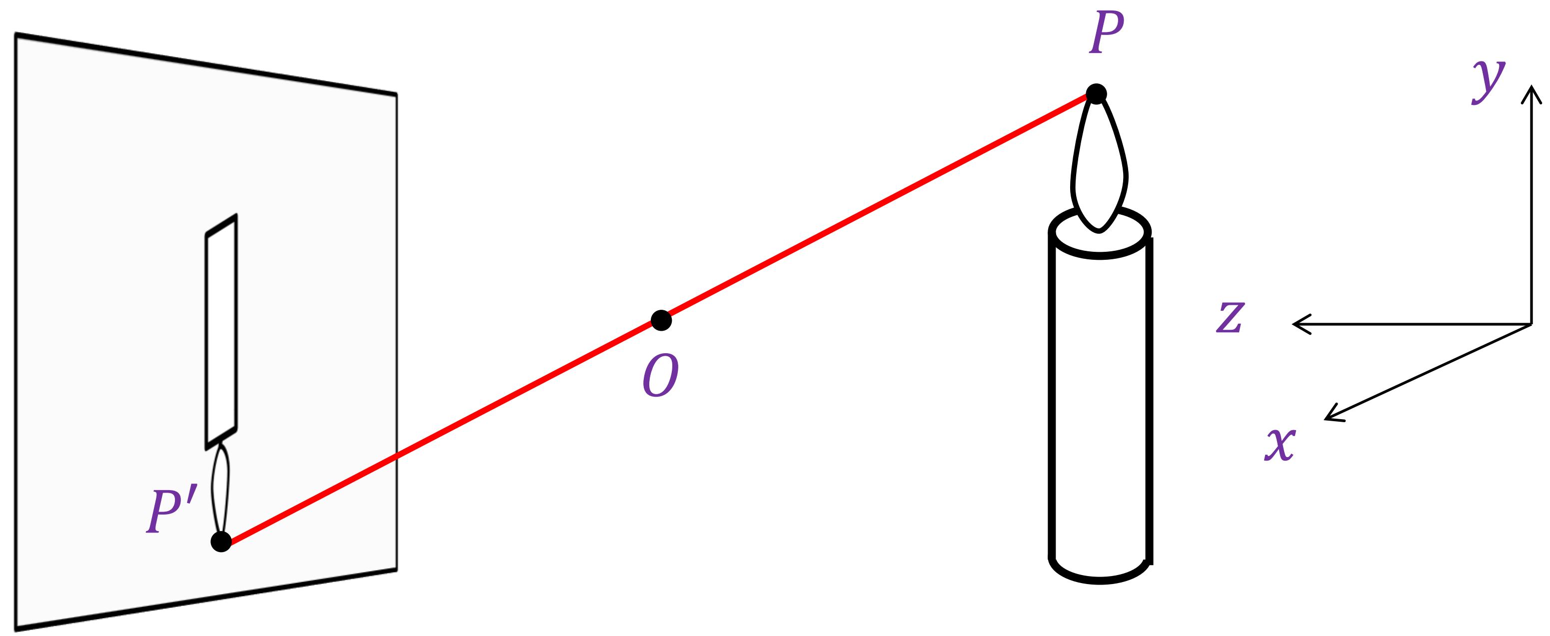
A. Torralba and W. Freeman. Accidental Pinhole and Pinspeck Cameras. CVPR 2012

Modeling Projection



- How do we find the projection P' of a scene point P?
  - Form the **visual ray** connecting P to the camera center O and find where it intersects the image plane
- All scene points that lie on this visual ray have the same projection in the image
- Are there scene points for which this projection is undefined?

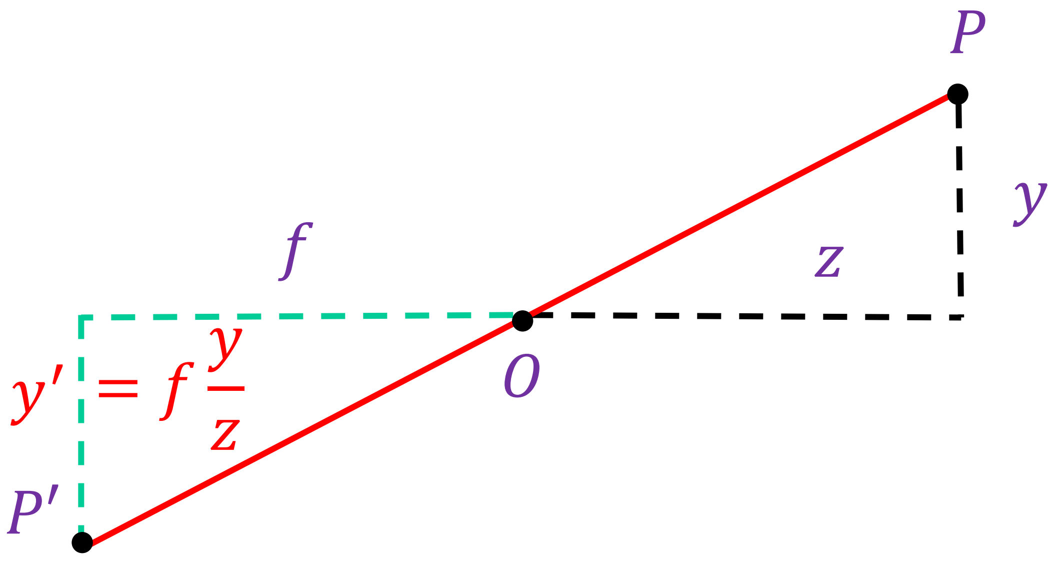
Modeling Projection



Canonical coordinate system

- The optical center (O) is at the origin
- The z axis is the *optical axis* perpendicular to the image plane
- The xy plane is parallel to the image plane, x and y axes are horizontal and vertical directions of the image plane

Deriving Projected Coordinates



$$\boxed{(x, y, z) \rightarrow \left(f\frac{x}{z}, f\frac{y}{z}\right)}$$

Properties of Projection

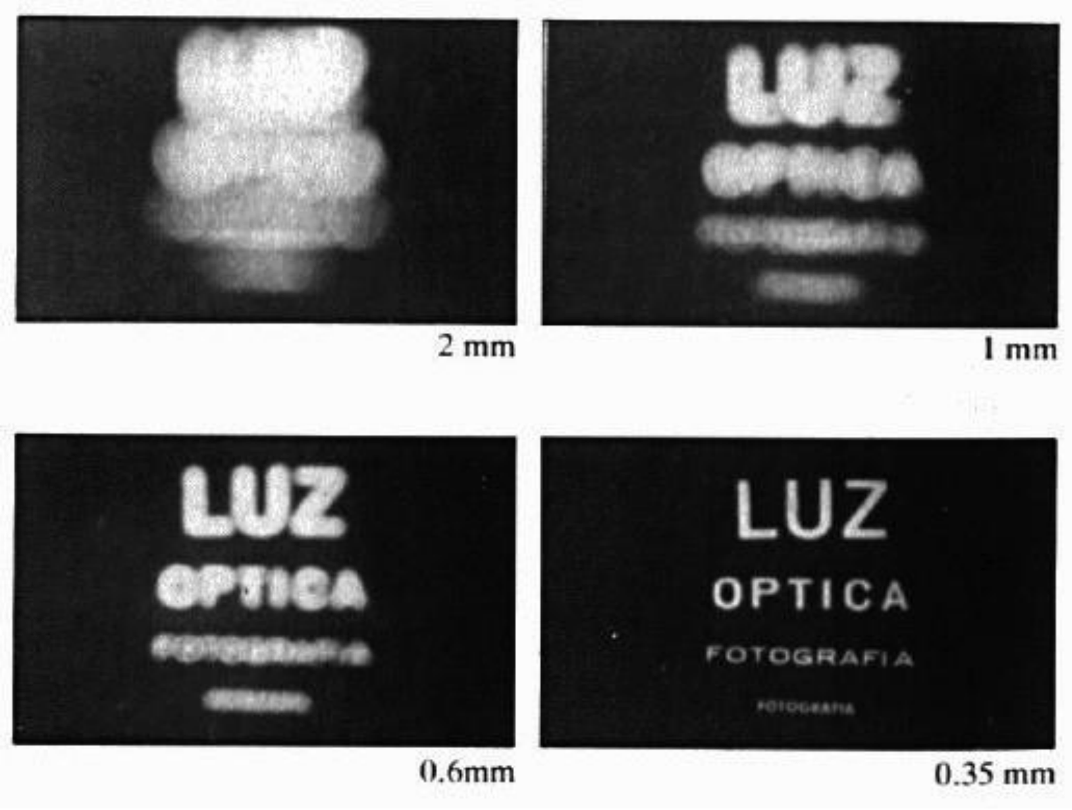
- Real-world sizes (lengths) are *not* preserved in projection
  - What other properties are/are not preserved?

Piero della Francesca, *Flagellation of Christ*, 1455-1460

Home-made Pinhole Camera

Source: P. Debevec via A. Efros

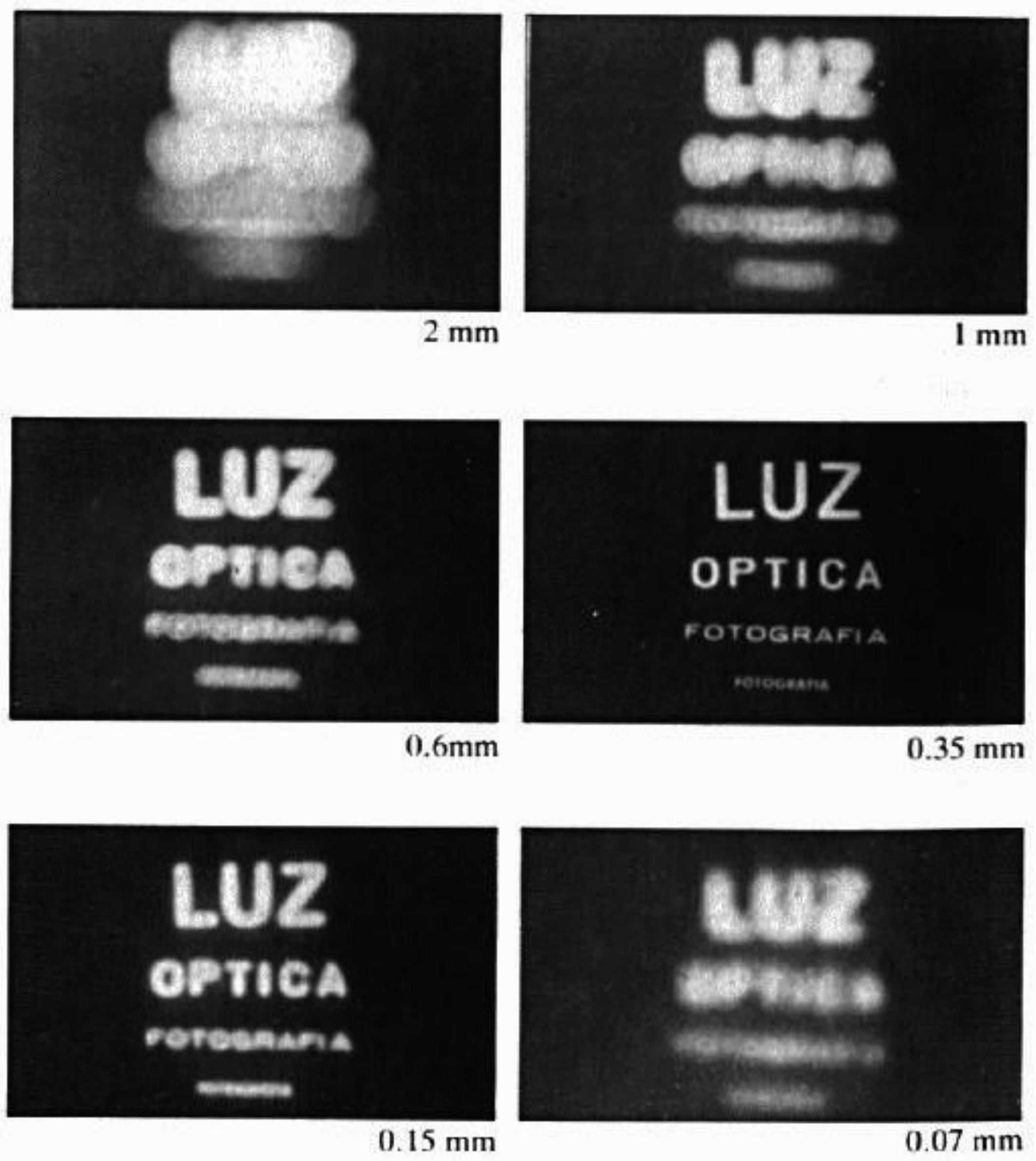
Shrinking the Aperture



Why not make the aperture as small as possible?

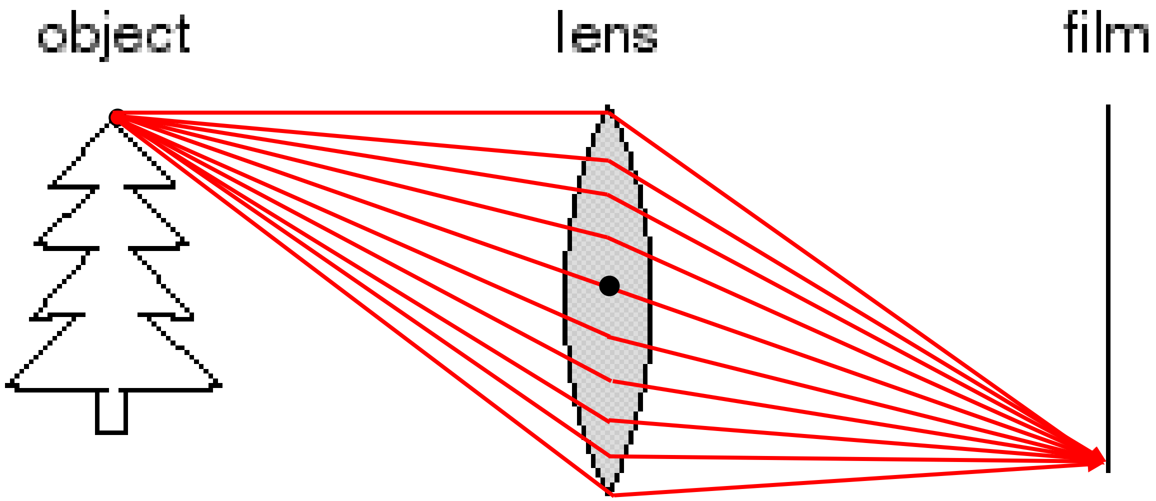
- Less light gets through
- Diffraction!

Shrinking the Aperture



Lenses

Adding a Lens

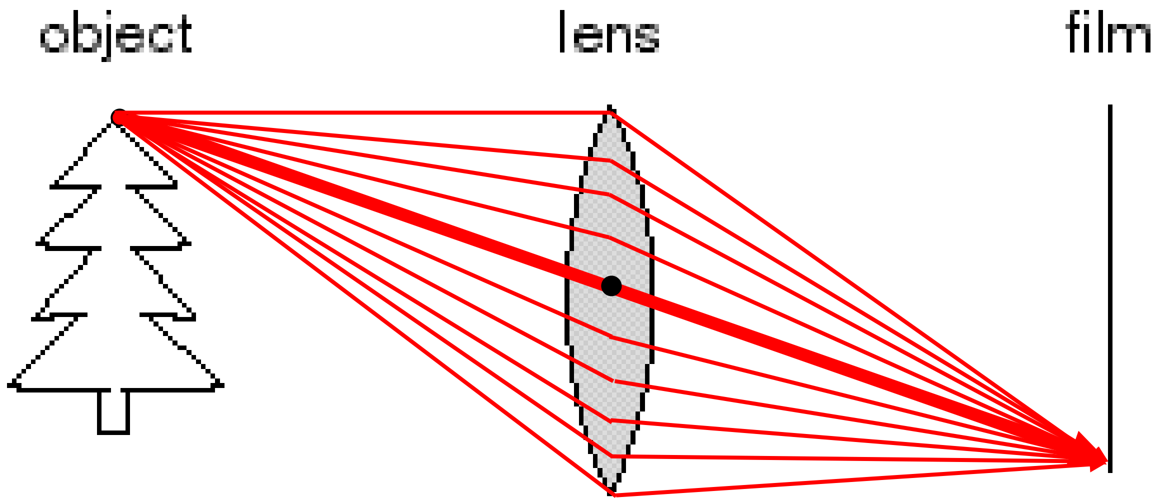


A lens focuses light onto the film

- Thin lens model:

Slide by Steve Seitz

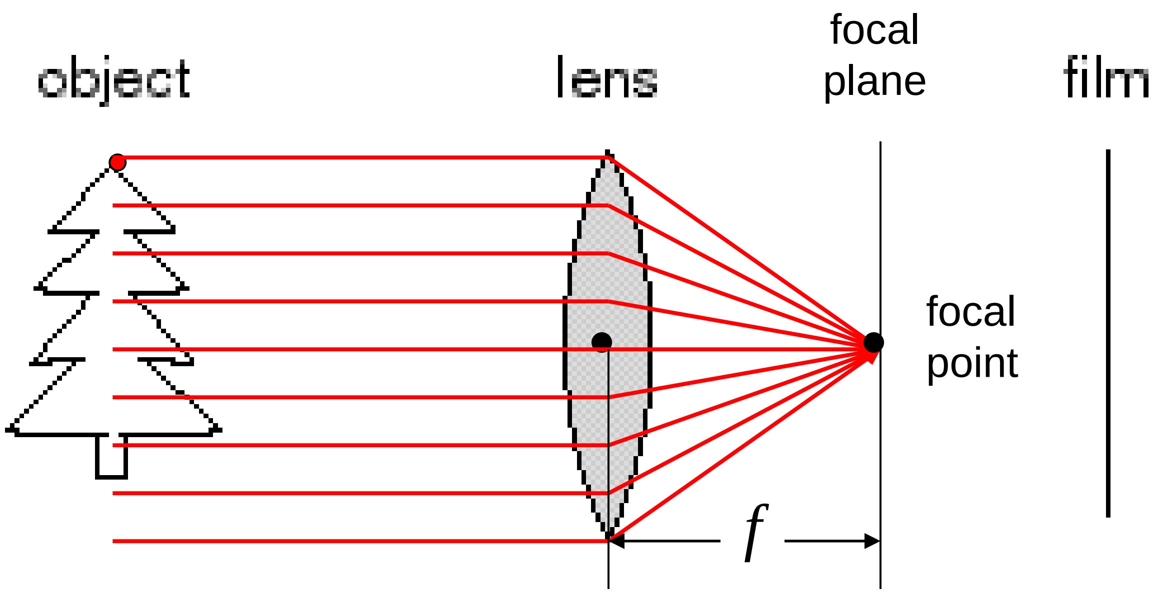
Adding a Lens



A lens focuses light onto the film

- Thin lens model:
  - Rays passing through the center are not deviated (pinhole projection model still holds)

Adding a Lens



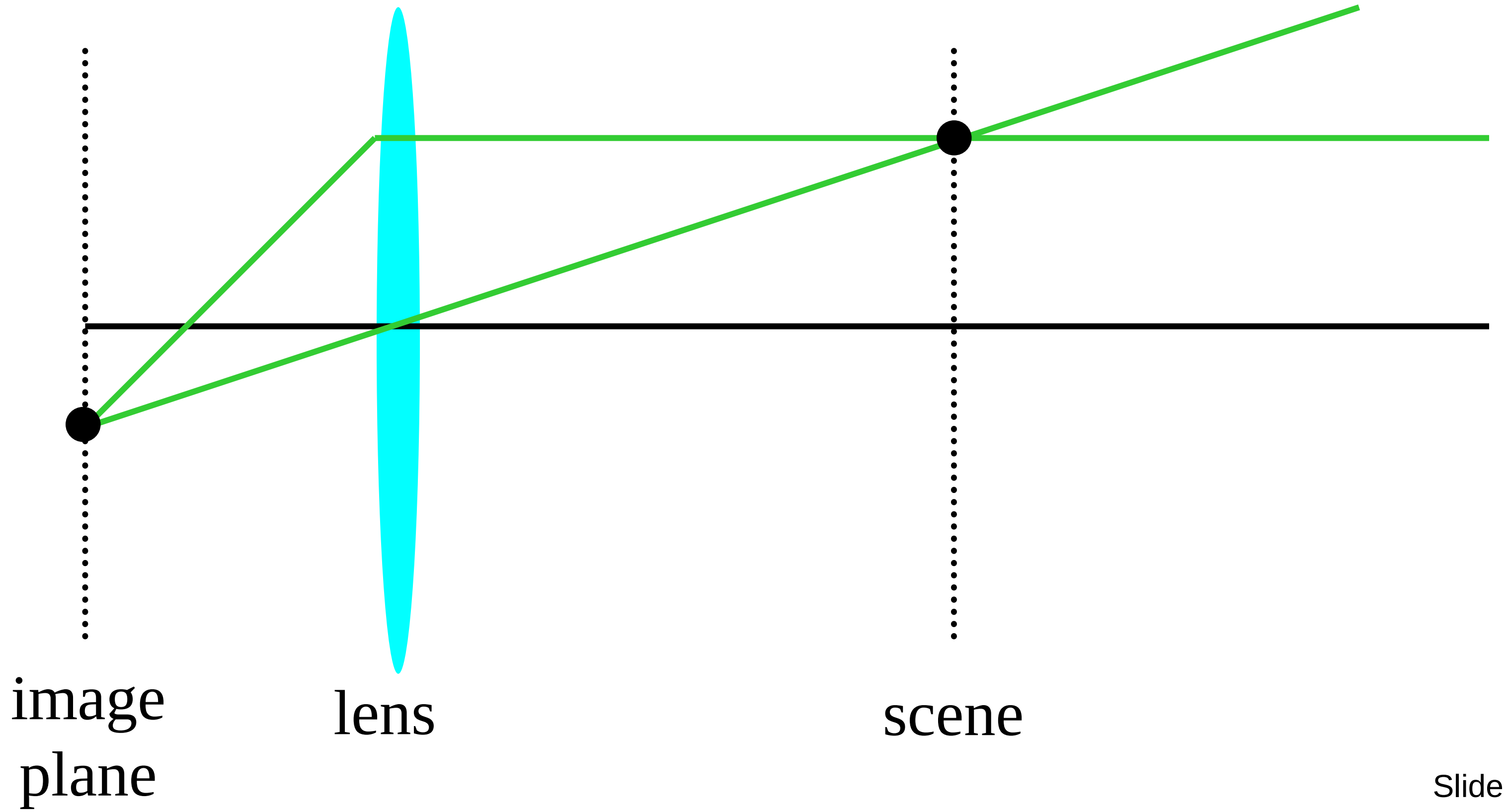
A lens focuses light onto the film

- Thin lens model:
  - Rays passing through the center are not deviated (pinhole projection model still holds)
  - All rays parallel to the optical axis pass through the *focal point*
  - All parallel rays converge to points on the *focal plane*

Slide by Steve Seitz

Thin Lens Formula

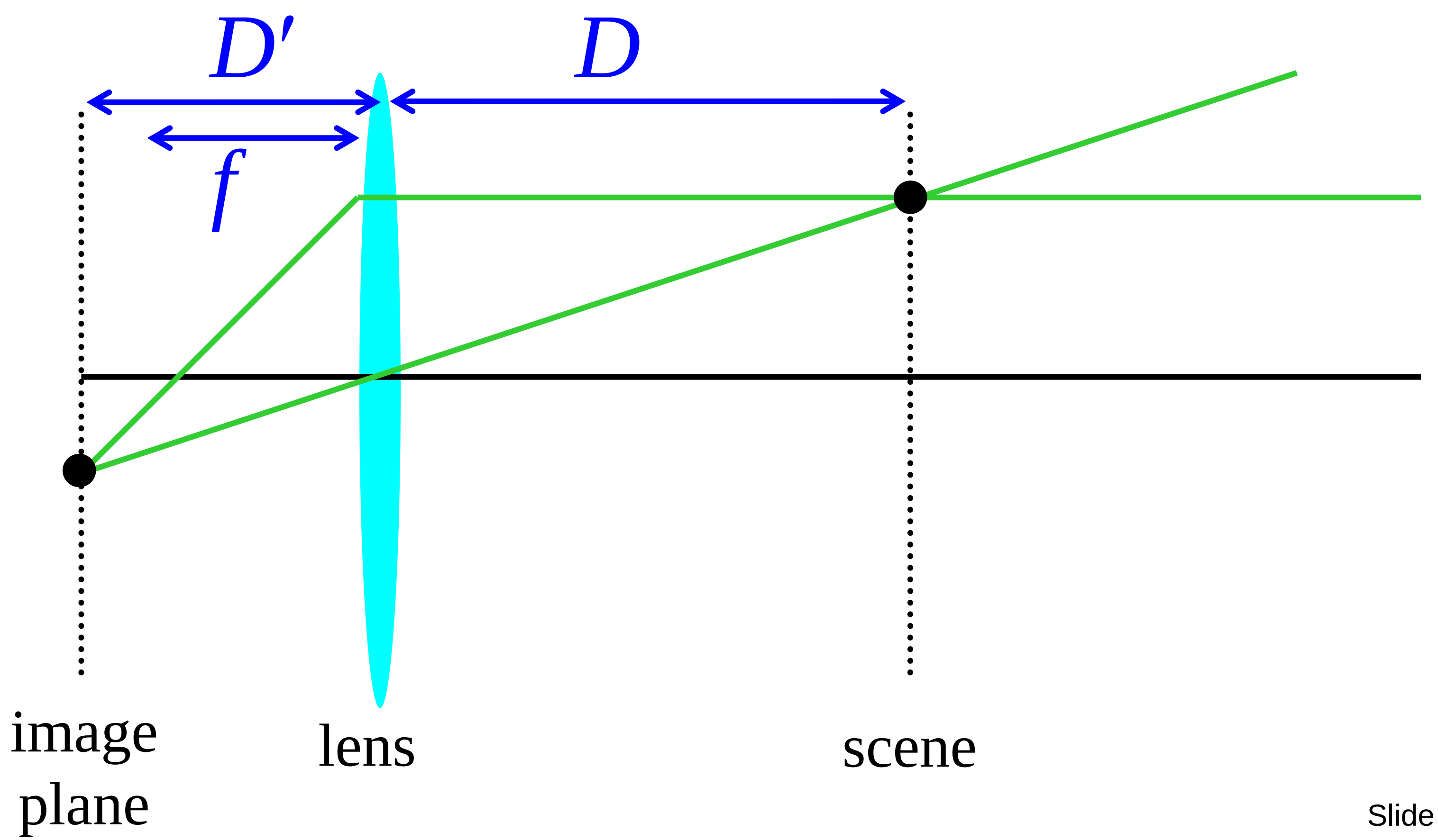
- Where does the lens focus the rays coming from a given point in the scene?



Slide by Frédo Durand

Thin Lens Formula

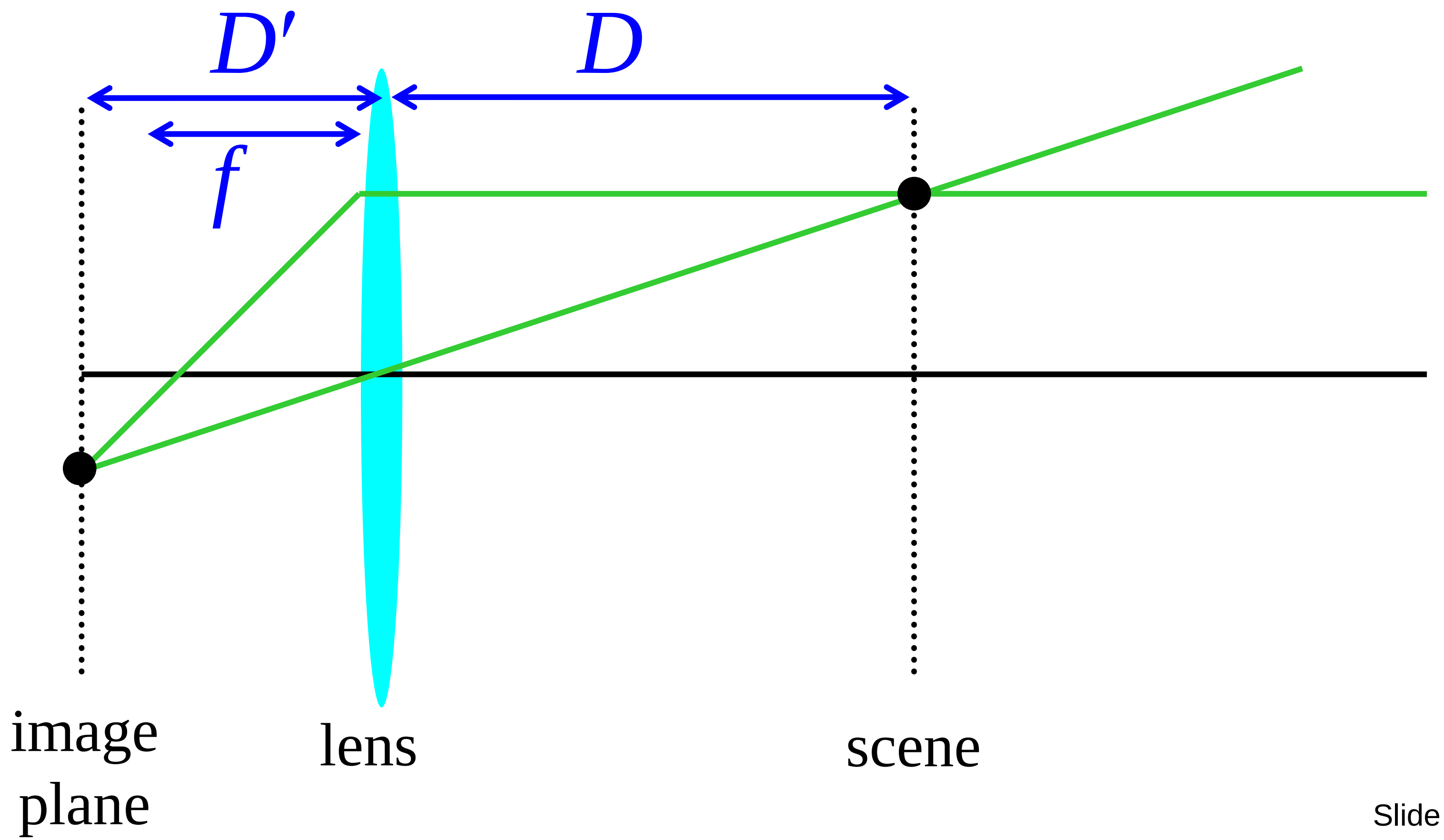
- What is the relation between the focal length (f), the distance of the object from the optical center (D), and the distance at which the object will be in focus (D')?



Slide by Frédo Durand

Thin Lens Formula

Similar triangles everywhere!

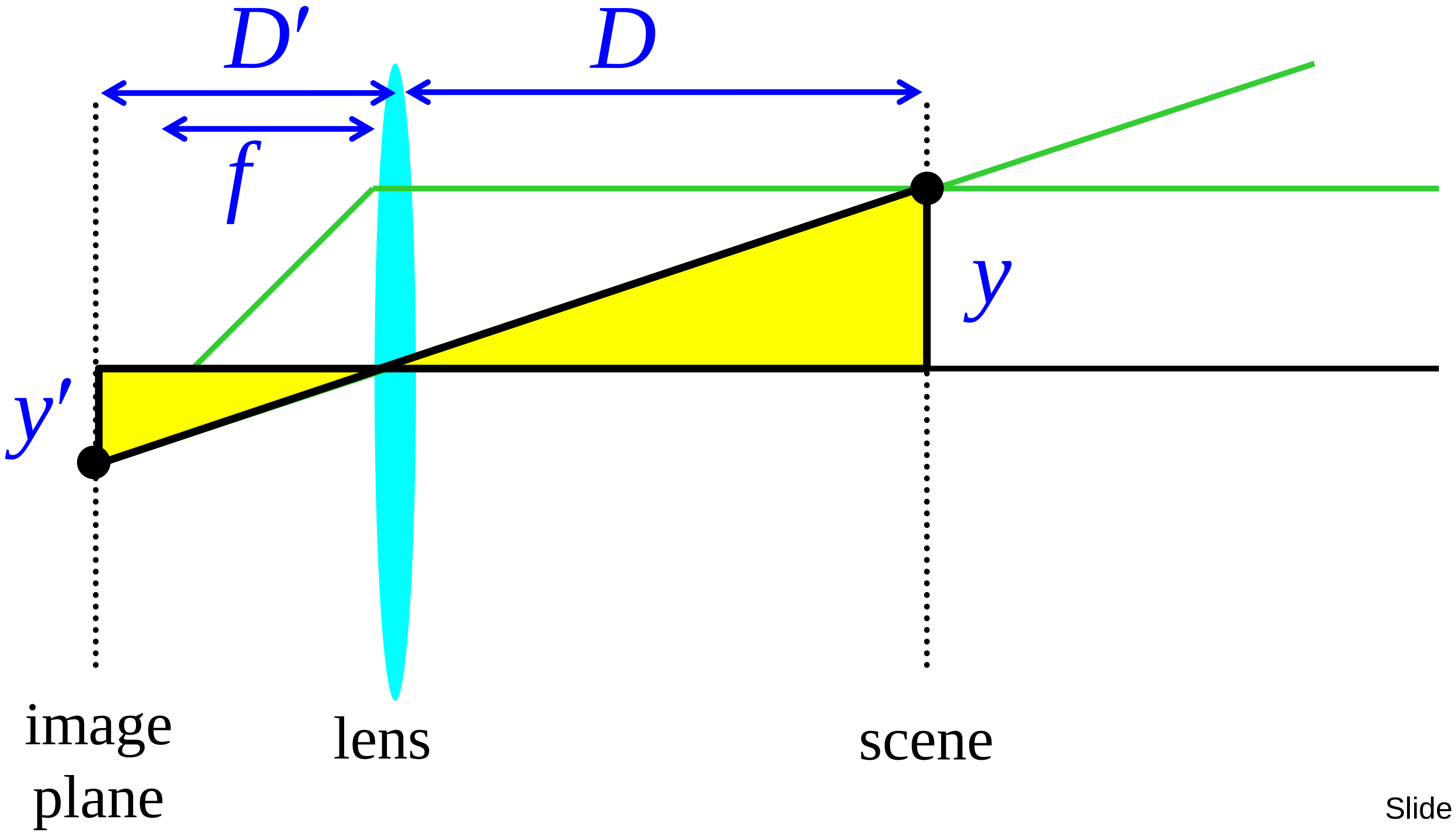


Slide by Frédo Durand

Thin Lens Formula

Similar triangles everywhere!

$y'/y = D'/D$



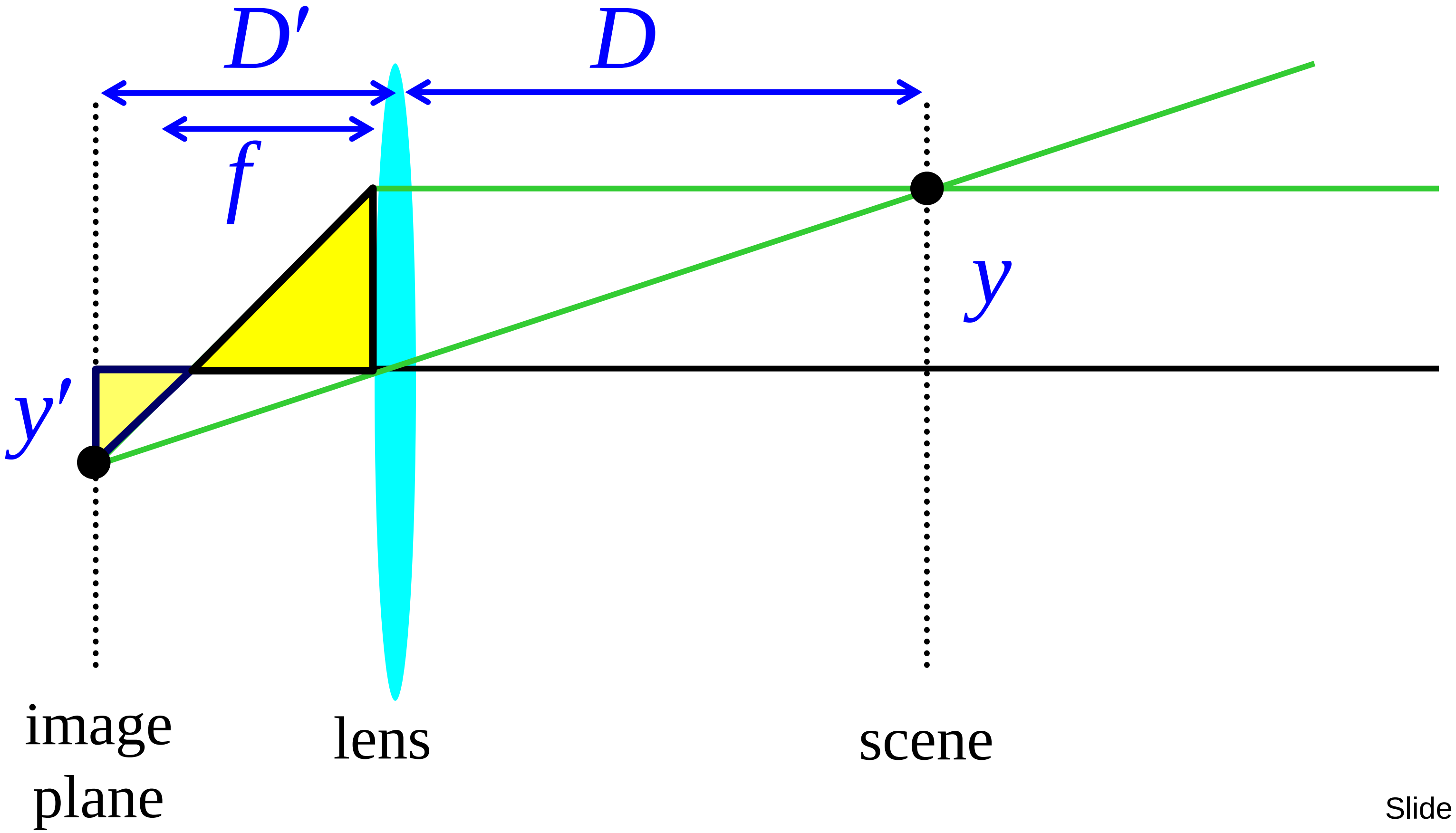
Slide by Frédo Durand

Thin Lens Formula

Similar triangles everywhere!

$$y'/y = D'/D$$

$$y'/y = (D'-f)/f$$



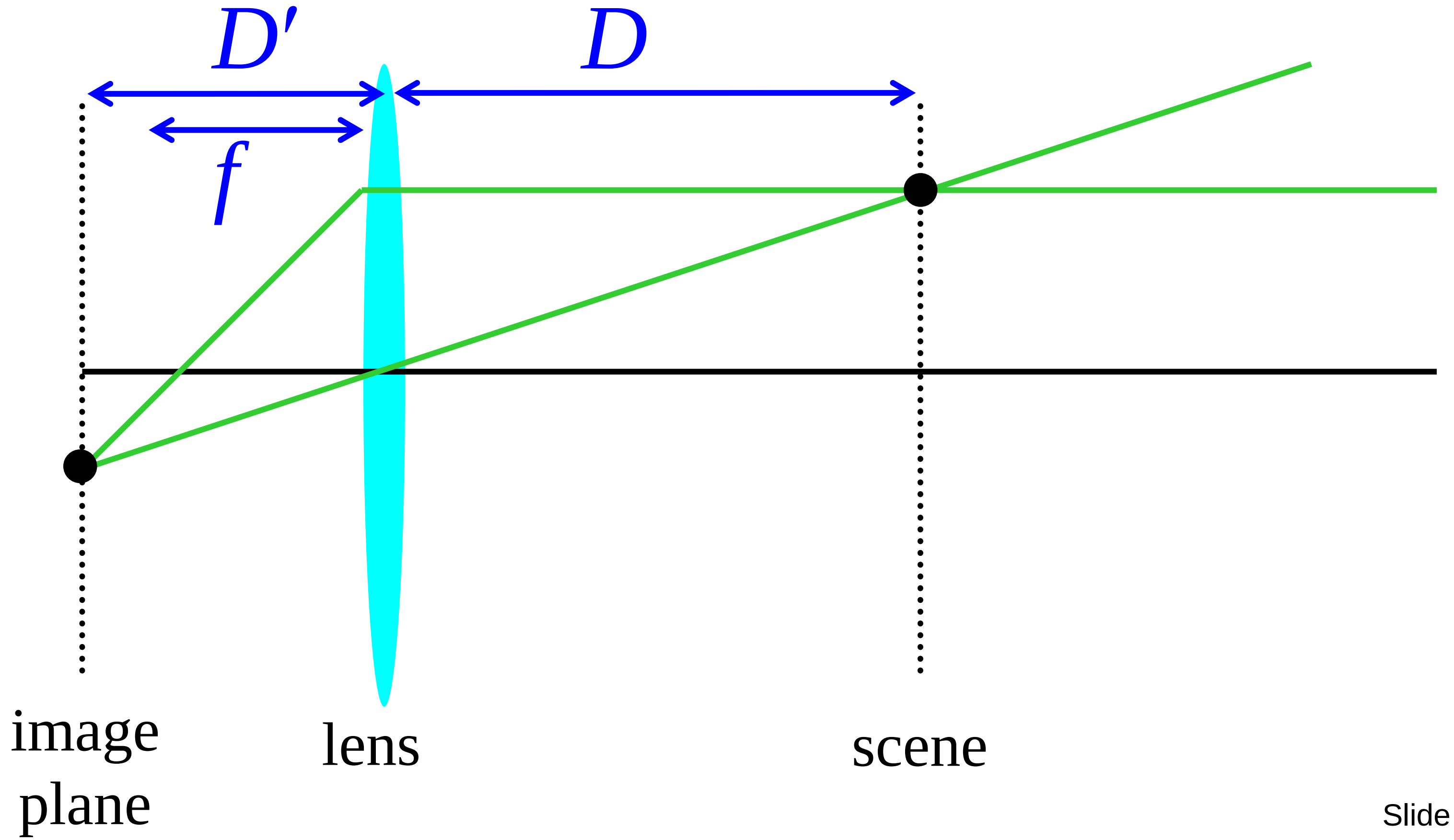
Slide by Frédo Durand

Thin Lens Formula

$$\boxed{\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}}$$

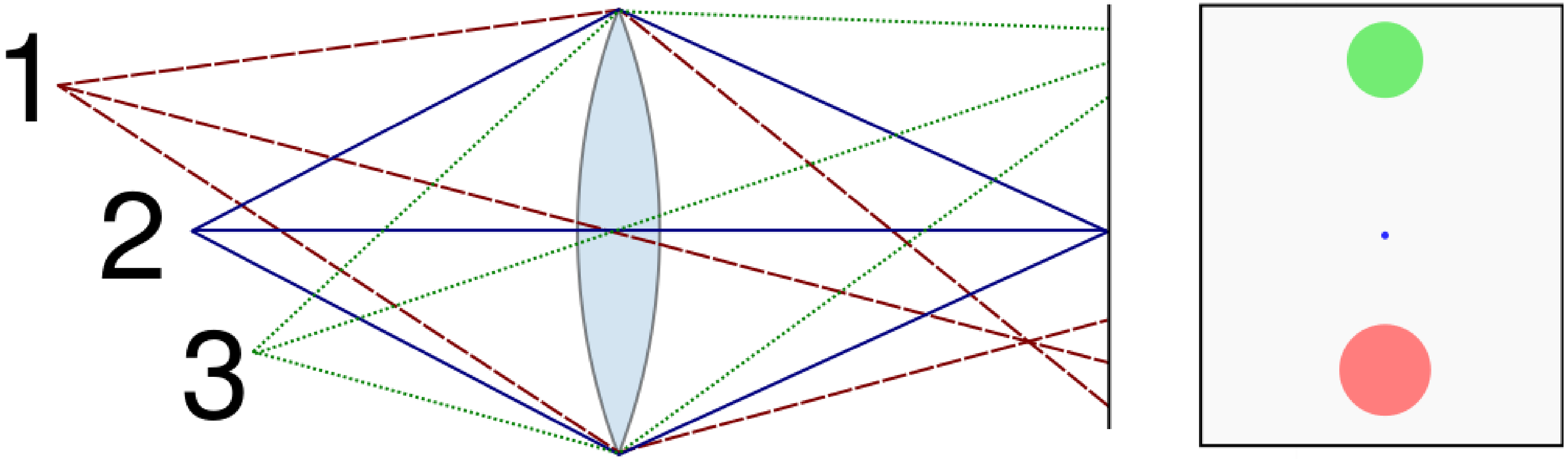
Any point satisfying the thin lens equation is in focus.

What happens when D is very large?



Slide by Frédo Durand

Depth of field



- For a fixed focal length and image plane, there is a specific distance at which objects are “in focus”
  - Other points project to a “circle of confusion” in the image

Depth of Field

- Depth of field is the distance between the nearest and farthest objects in a scene that appear acceptably sharp in an image (Wikipedia)

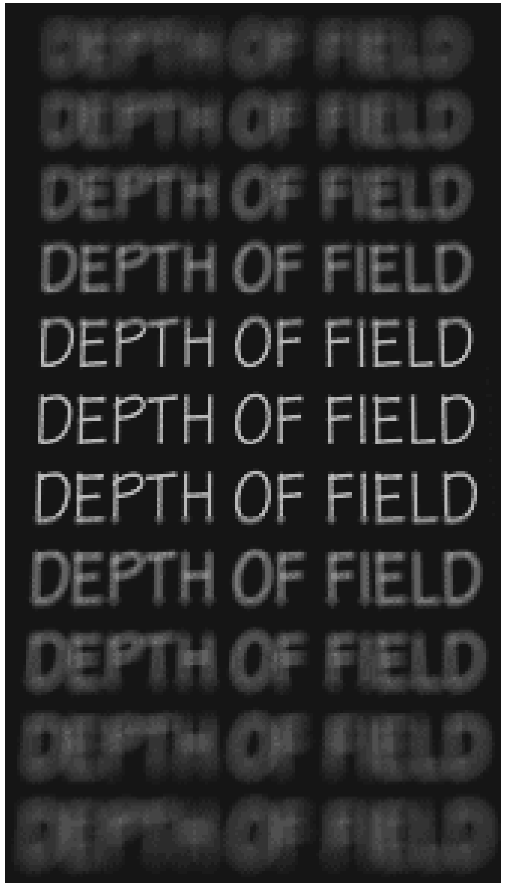


Image source
(via A. Efros)

Controlling Depth of Field

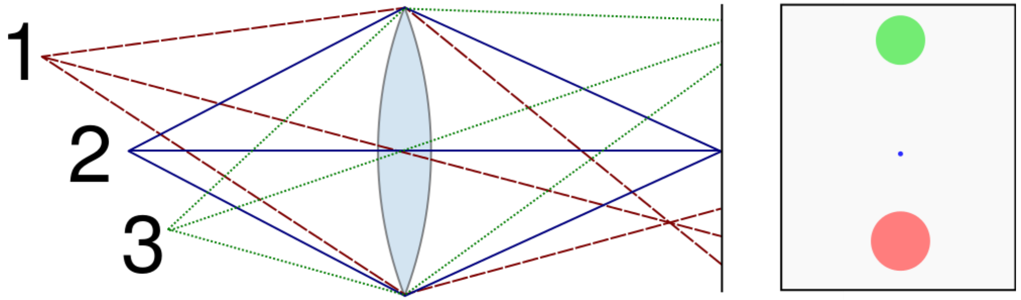
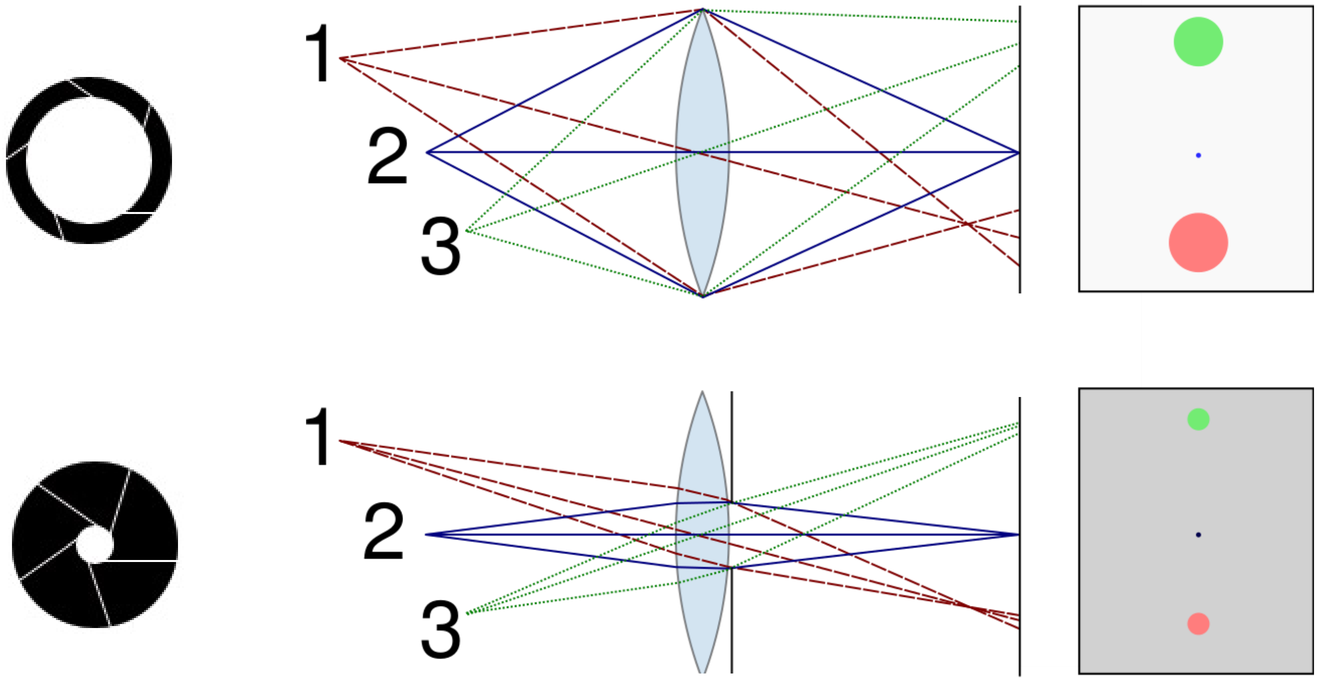


Figure source

Controlling Depth of Field



Changing the *aperture* size affects depth of field

- A smaller aperture increases the range in which the object is approximately in focus
- But small aperture reduces amount of light – need to increase *exposure*

Figure source

Varying the Aperture

Large aperture = small DOF

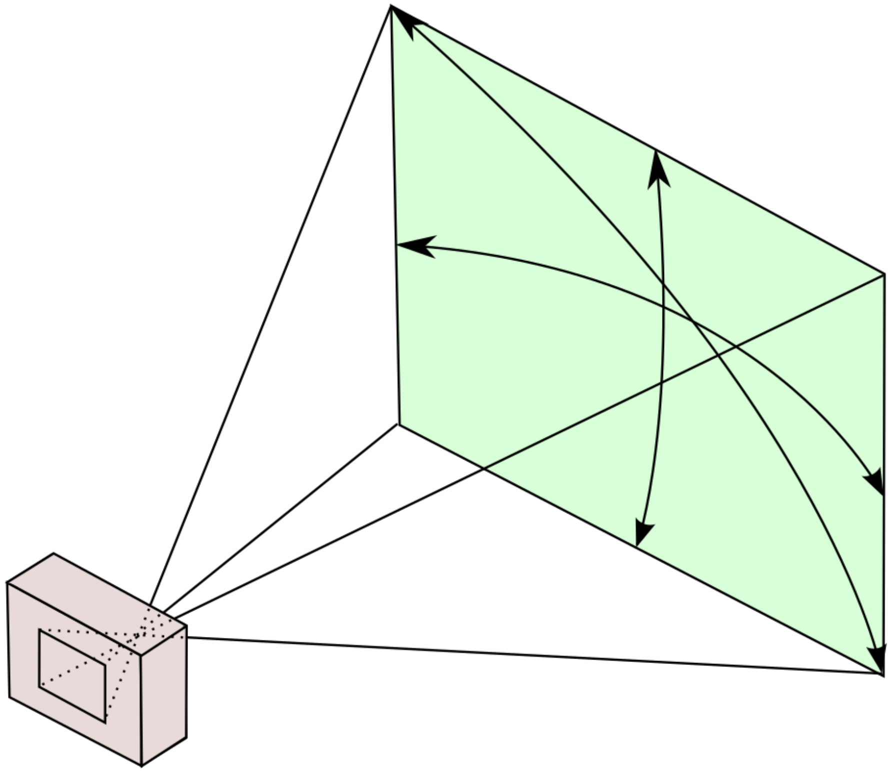
Small aperture = large DOF

Slide by A. Efros

Field of View, Focal Length

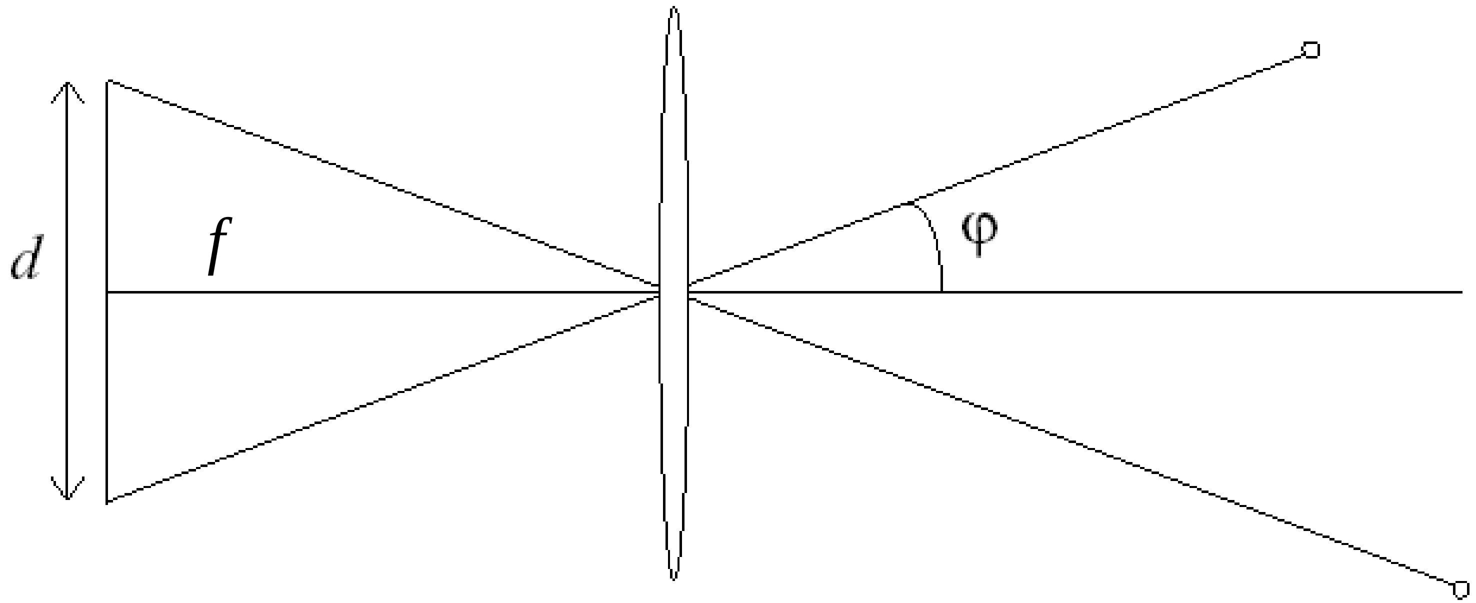
Field of View

- The field of view is the angular extent of the world observed by the camera (Wikipedia)
- What determines the FOV?



Field of View

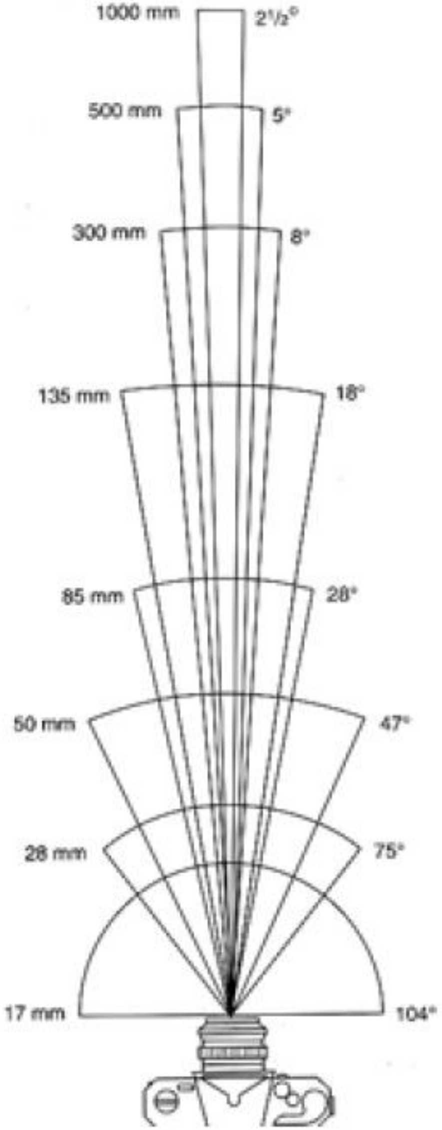
- The field of view is the angular extent of the world observed by the camera (Wikipedia)
- What determines the FOV?
  - Focal length (f), length of the sensor (d):



$$\varphi = \tan^{-1}\frac{d}{2f}$$

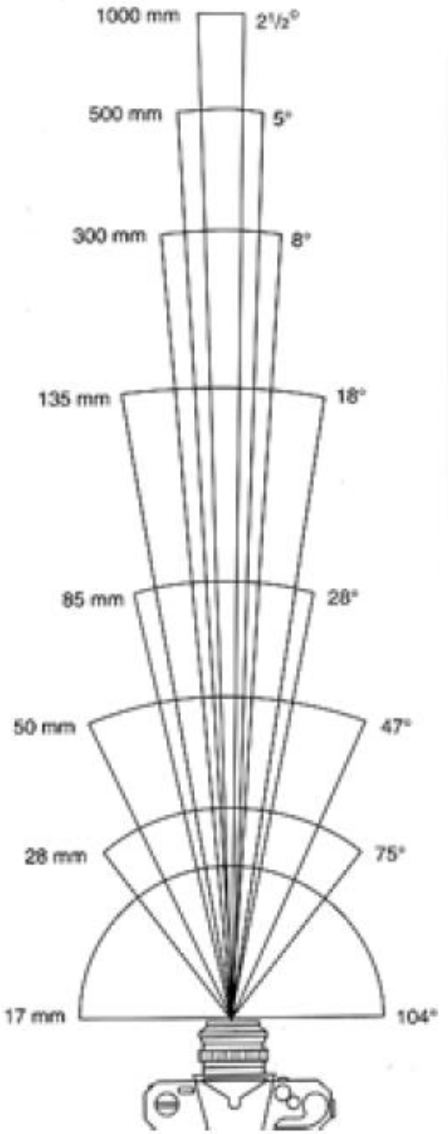
  - Larger focal length = smaller FOV

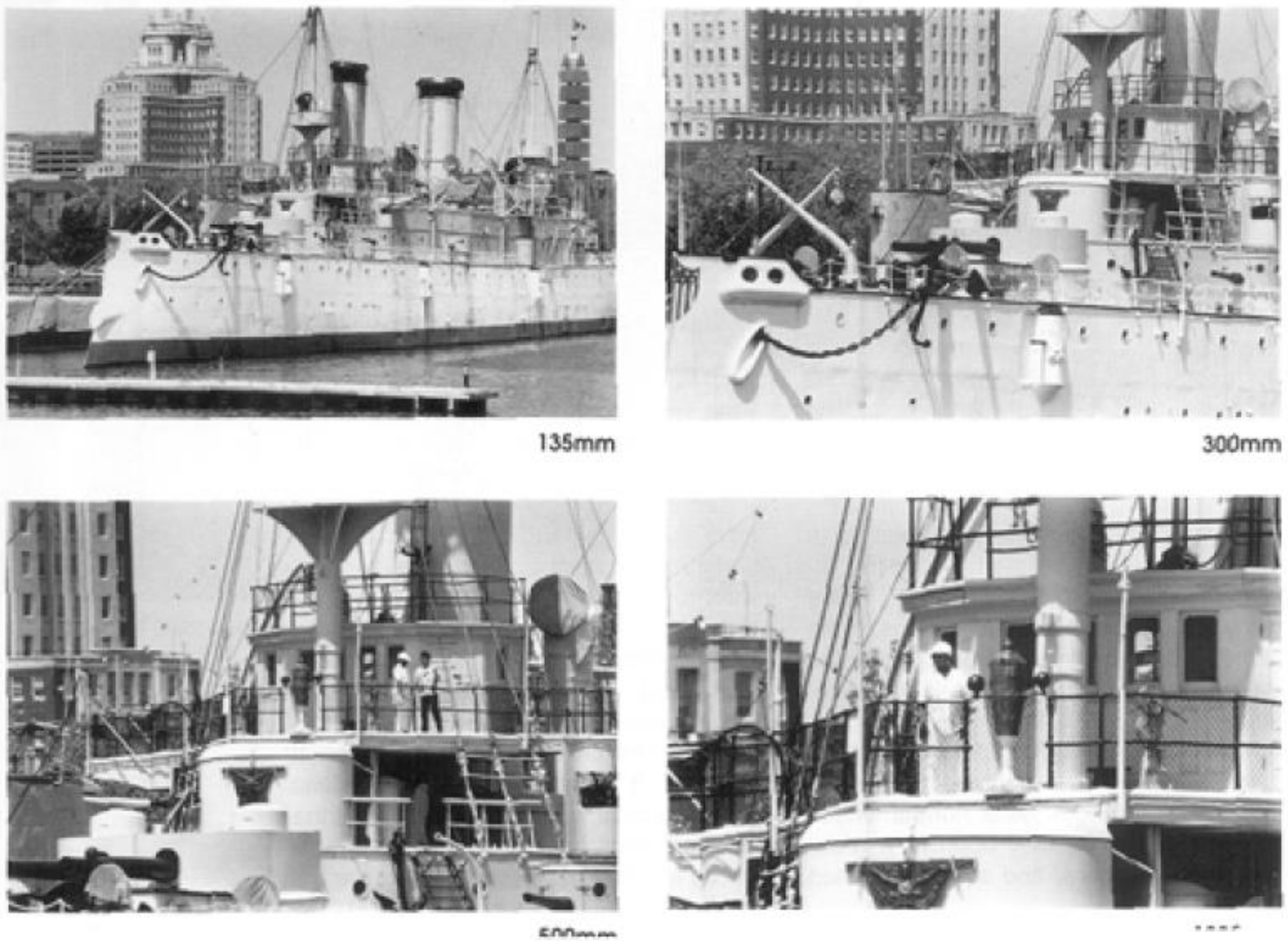
Field of View



Slide by A. Efros

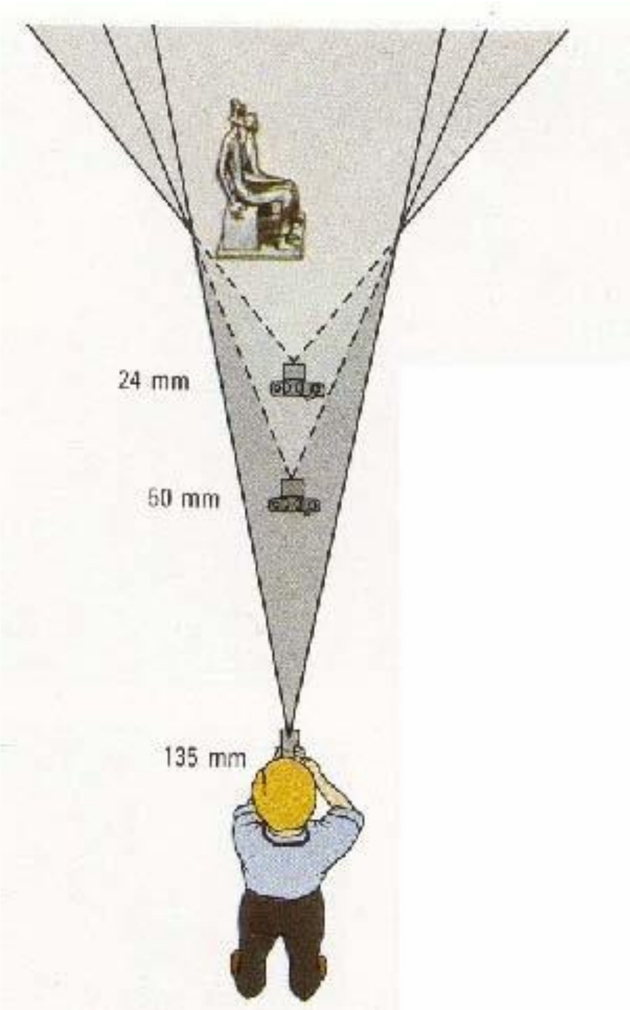
Field of View





Slide by A. Efros

Field of View / Focal Length



Large FOV, small *f*
Camera close to car

Small FOV, large *f*
Camera far from the car

Sources: A. Efros, F. Durand

Same Effect for Faces

wide-angle

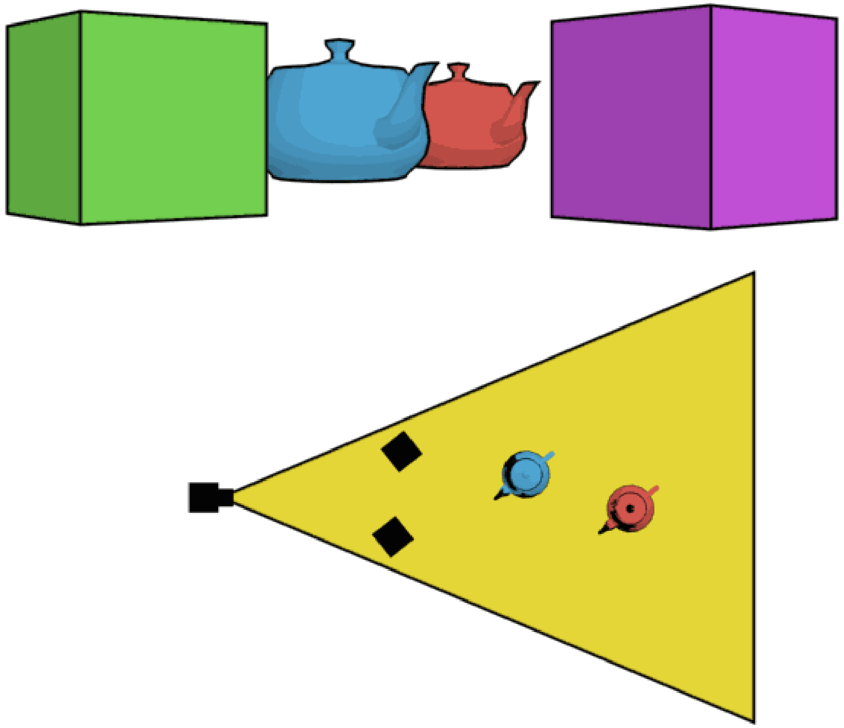
standard

telephoto

Source: F. Durand

The Dolly Zoom

- Continuously adjusting the focal length while the camera moves away from (or towards) the subject



http://en.wikipedia.org/wiki/Dolly_zoom

The Dolly Zoom

- Continuously adjusting the focal length while the camera moves away from (or towards) the subject
- “The Vertigo shot”

Example of dolly zoom from *Goodfellas* (YouTube)

Example of dolly zoom from *La Haine* (YouTube)

Another "Dolly Zoom"

Another "Dolly Zoom"

Choice of lens and viewpoint: A COVID-era illustration

Source

Marginal Distortion

Marginal Distortion

- In wide-angle photos, objects on the periphery “look wrong”

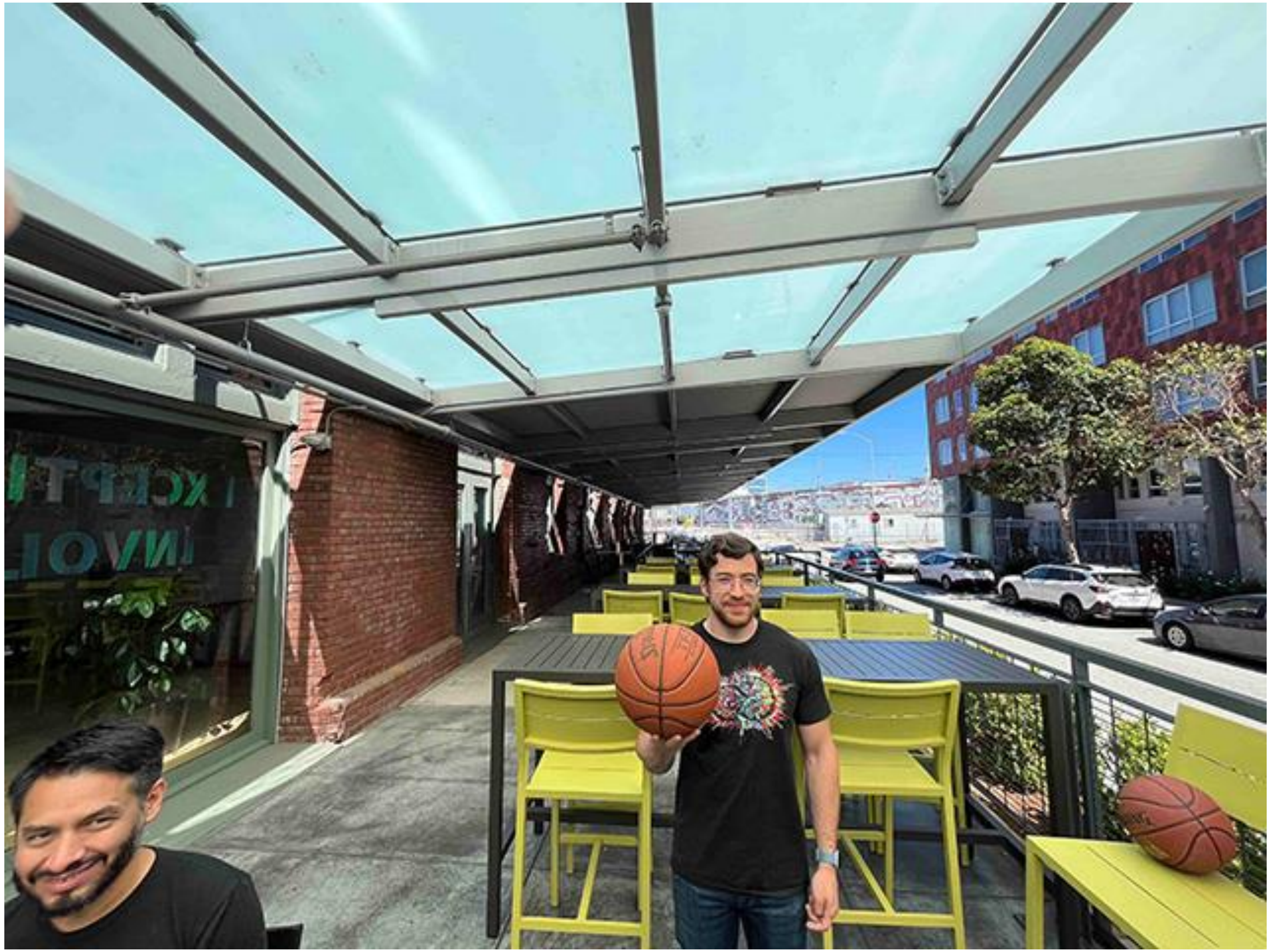


Image source:
A. Hertzmann

Marginal Distortion

- What is the shape of the projection of a sphere?

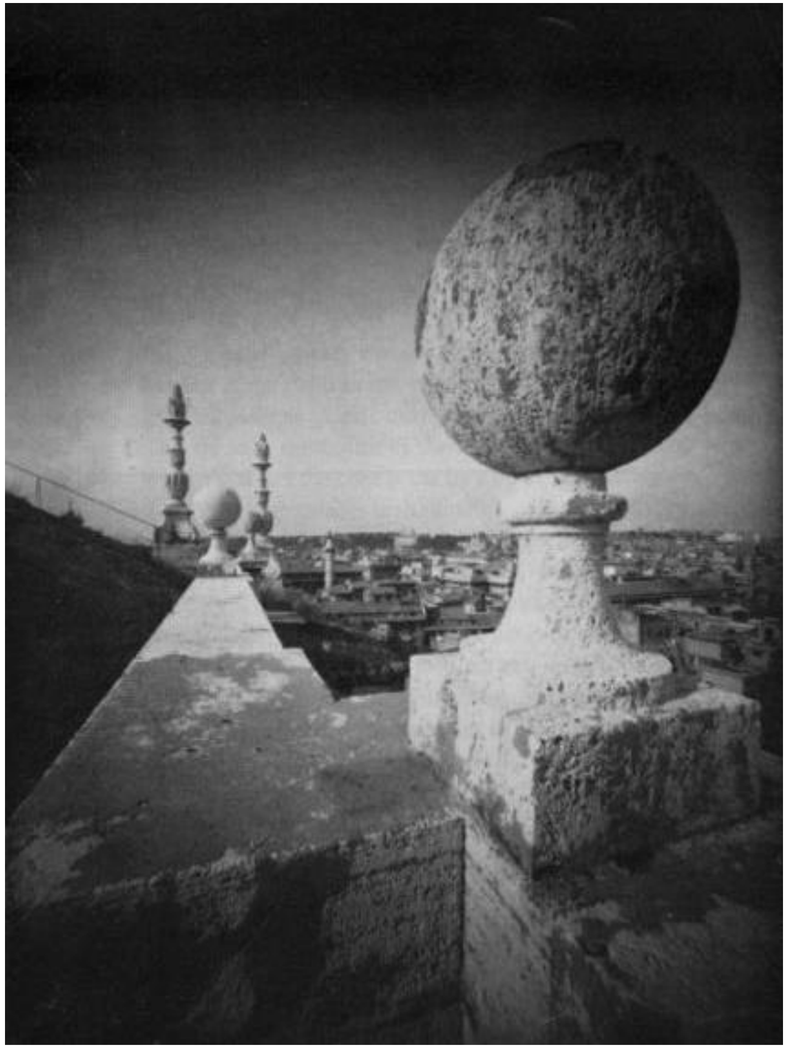


Image source: F. Durand

Marginal Distortion

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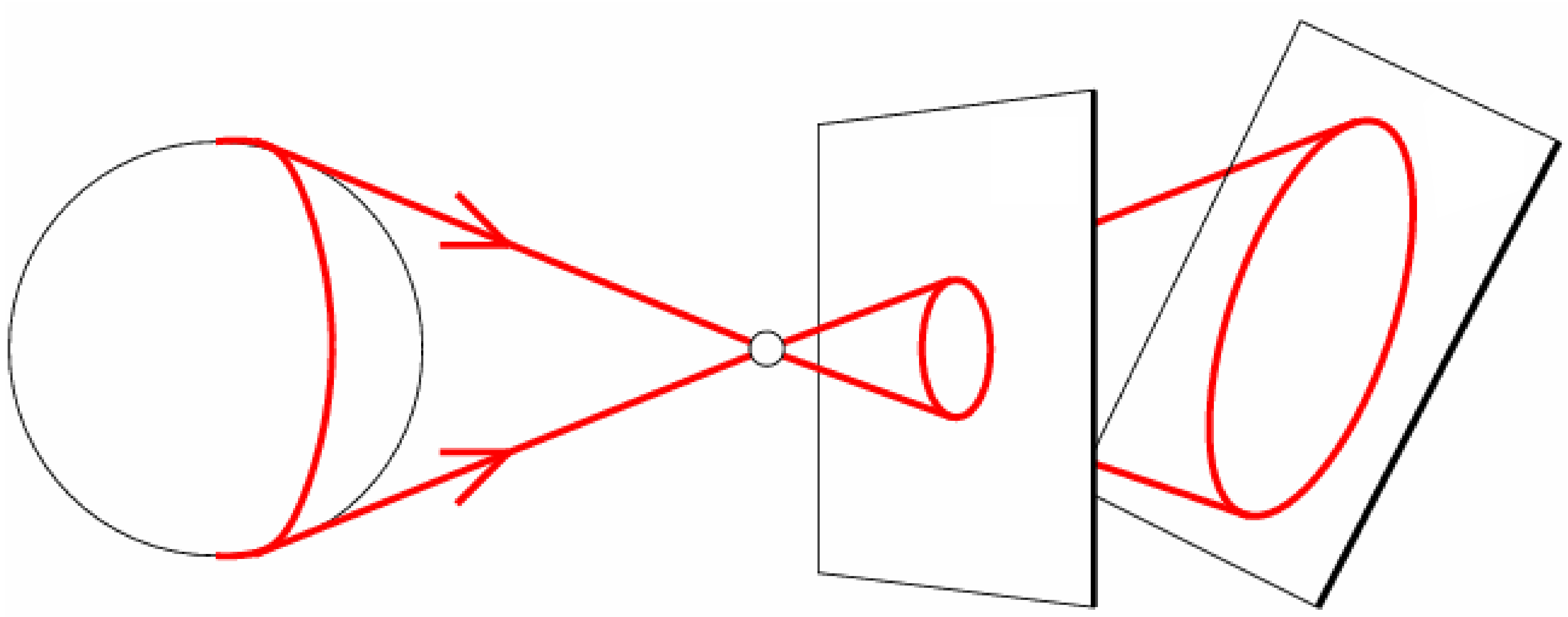
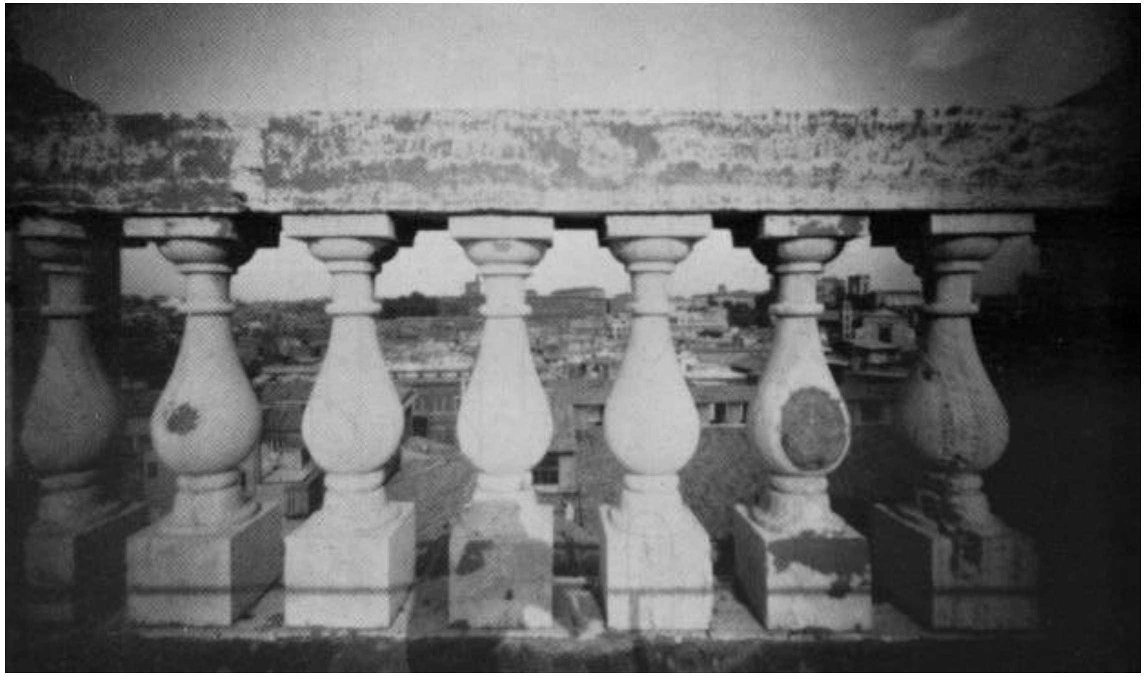
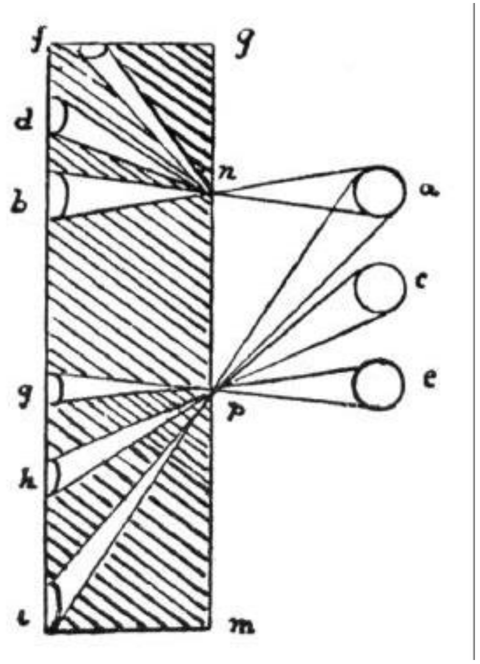


Marginal Distortion

- Are the widths of the projected columns equal?
  - The exterior columns are wider
  - This is not an optical illusion, and is not due to lens flaws
  - Phenomenon pointed out by Leonardo Da Vinci





Source: F. Durand

Marginal Distortion

- Which way are these people facing?

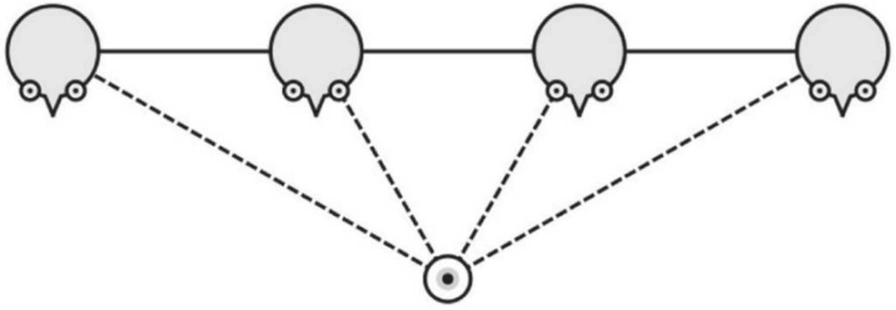
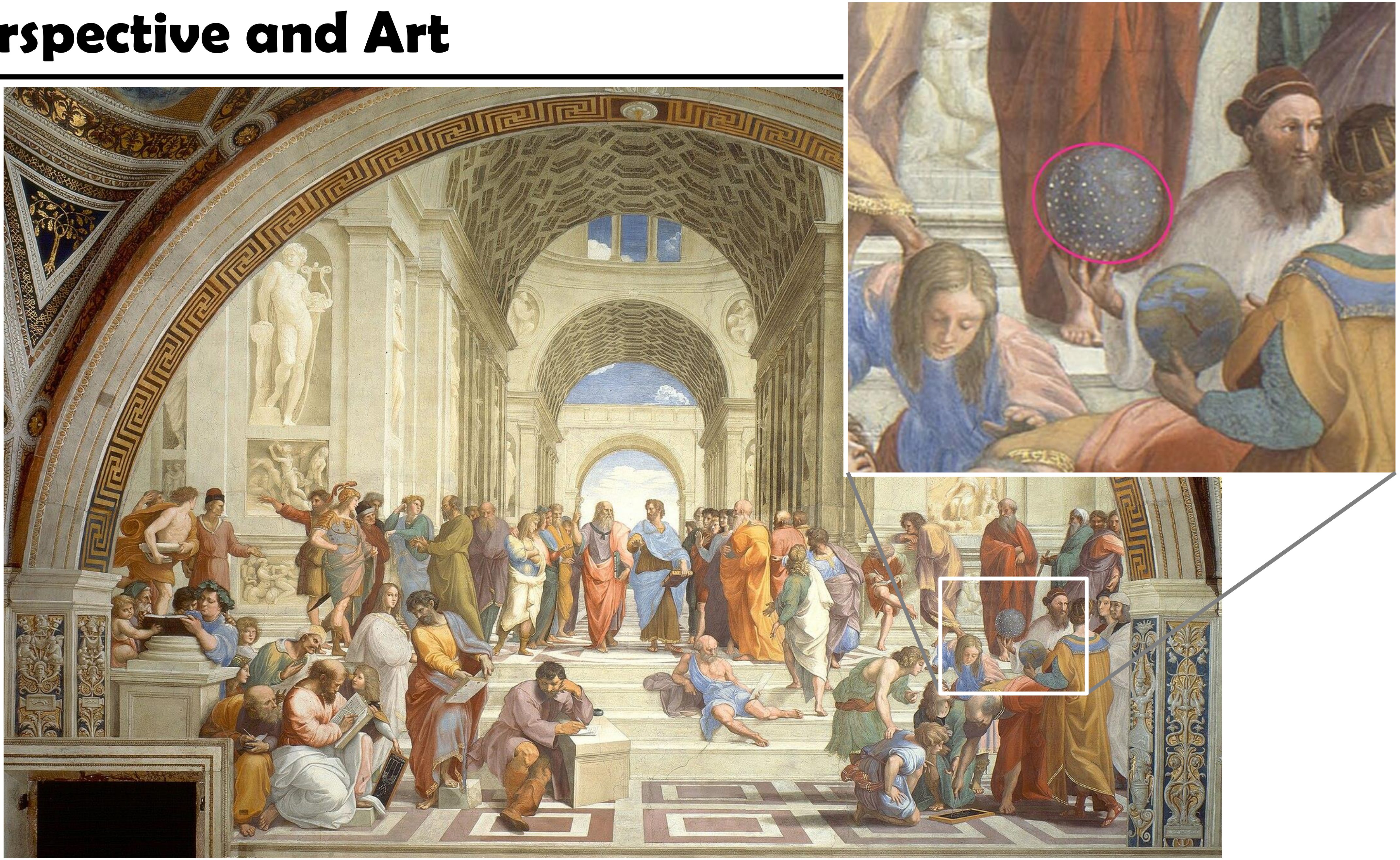


Image source: [Koenderink et al.]() via [A. Hertzmann]()

Perspective and Art



Raphael, The School of Athens, 1509-1511

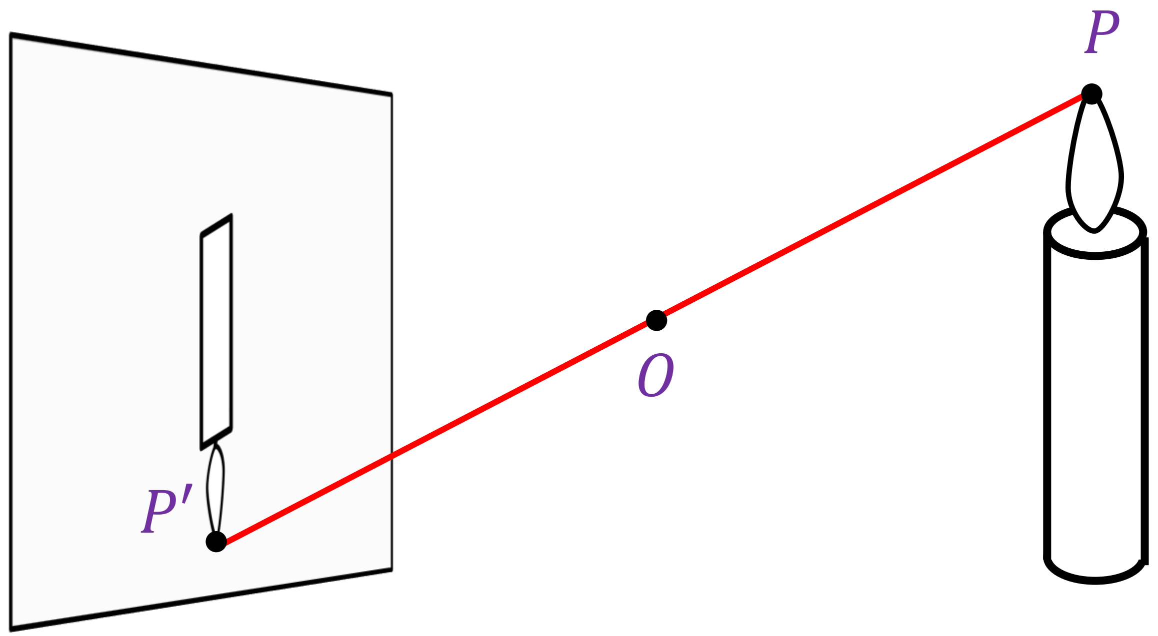
Perspective and Art

- “Large scenes with many faces are common in art… But, in the entire history of painting, I am unaware of any face depicted with the marginal distortions dictated by linear perspective.”

A. Hertzmann. [Toward a theory of perspective perception in pictures](). Journal of Vision, 2024

[Blog post]()

Review: Pinhole Projection



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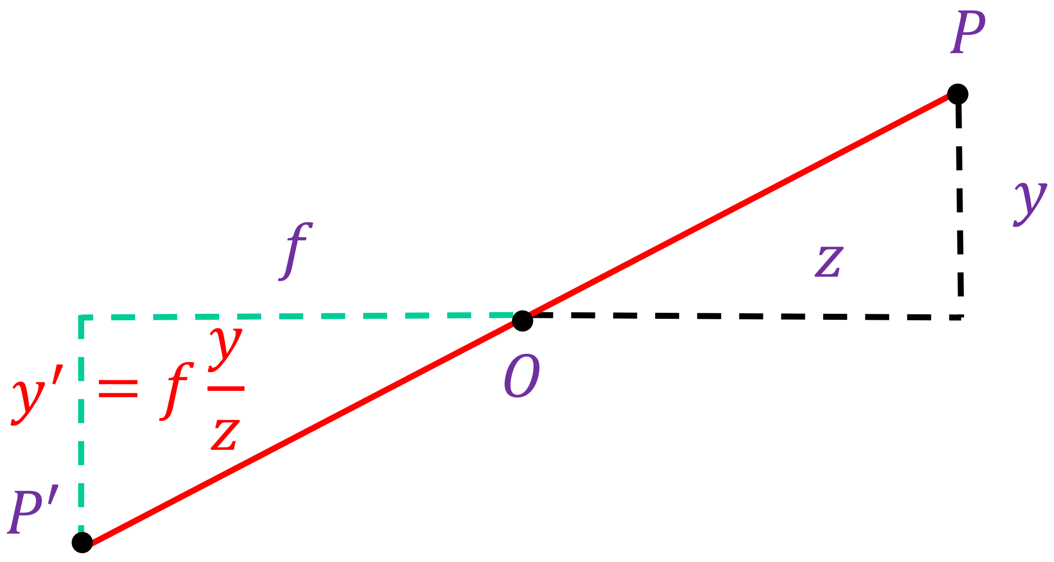
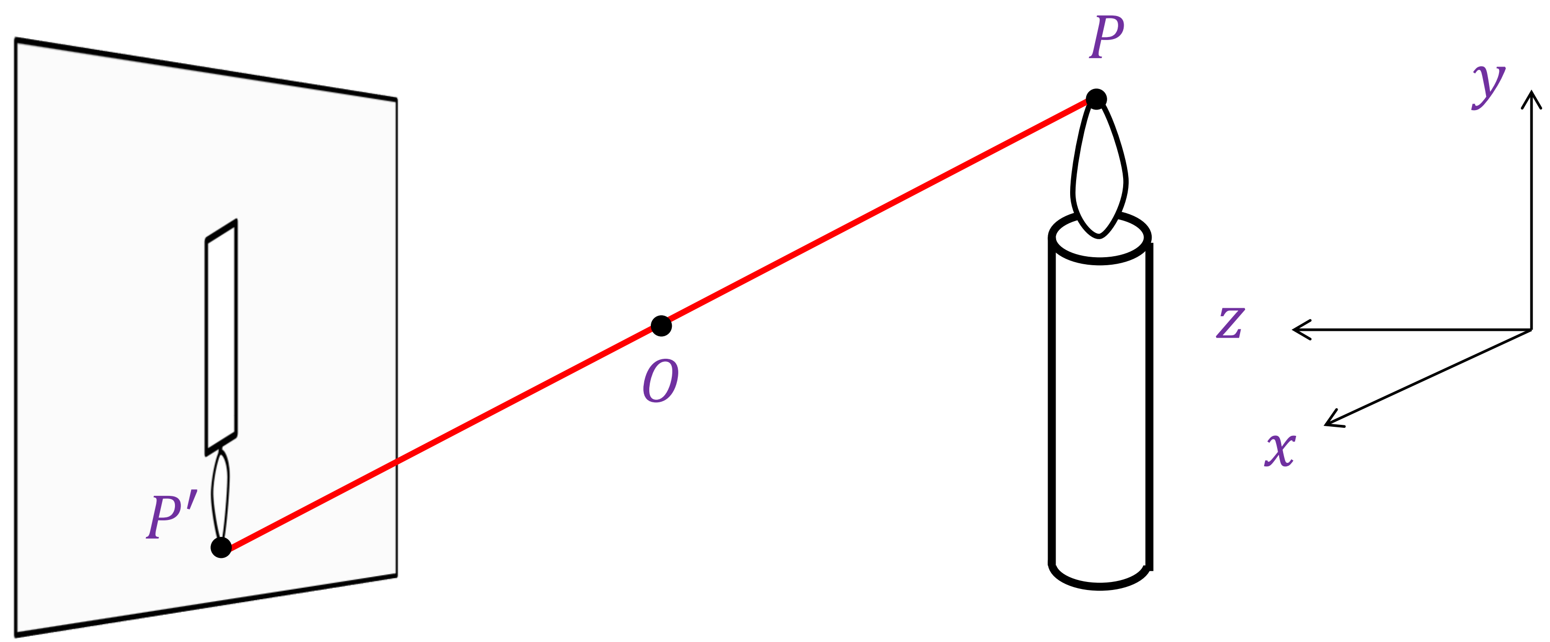


$$\boxed{(x, y, z) \rightarrow \left(f\frac{x}{z}, f\frac{y}{z}\right)}$$

Review: Pinhole Projection



Canonical coordinate system

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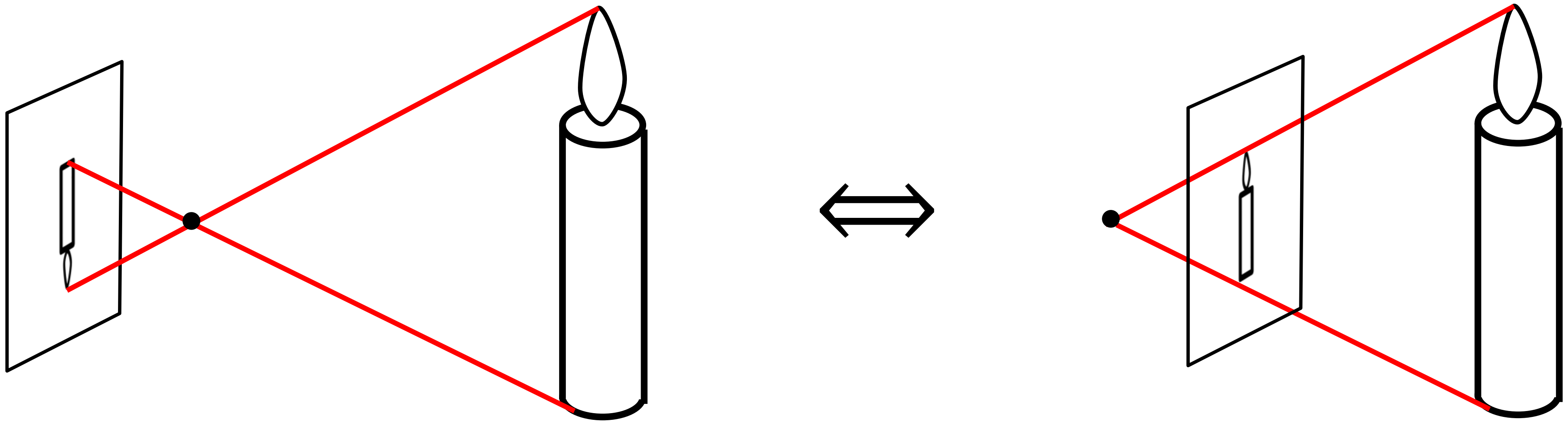
Review: Pinhole Projection



Note: instead of dealing with an image that is upside down, most of the time we will pretend that the image plane is *in front* of the camera center

Review: Field of View

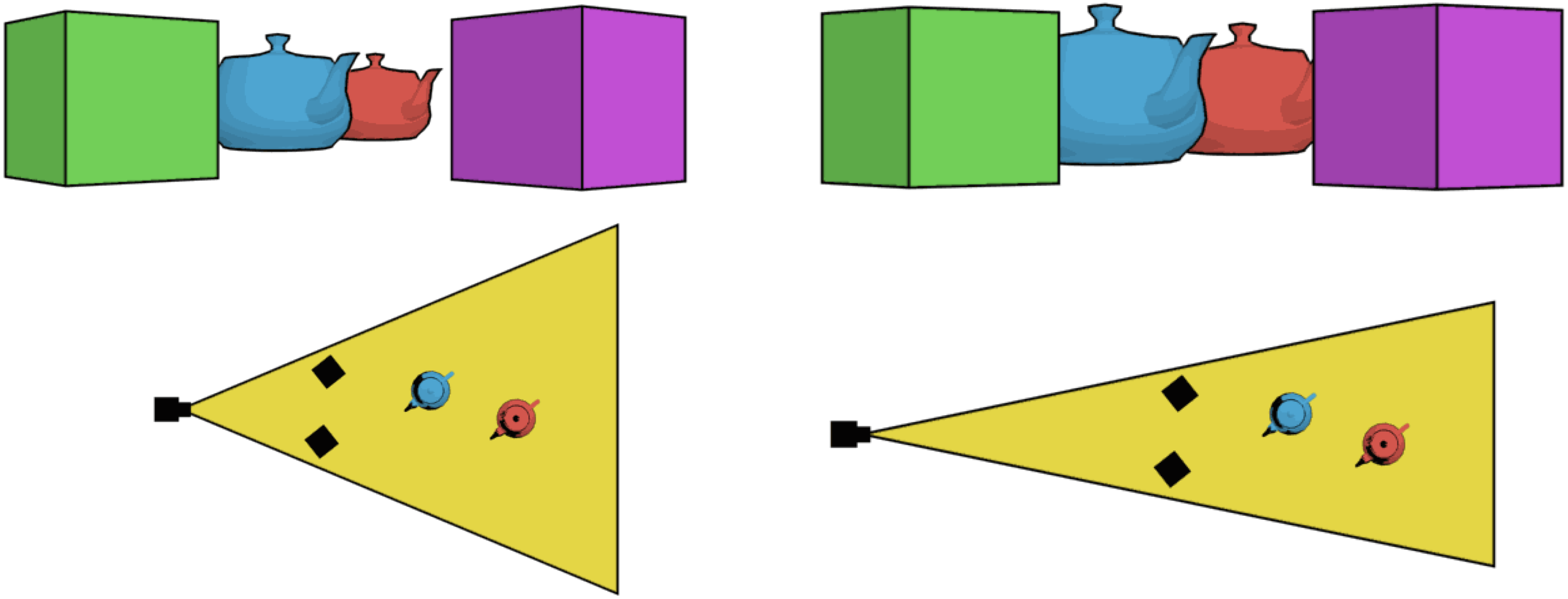
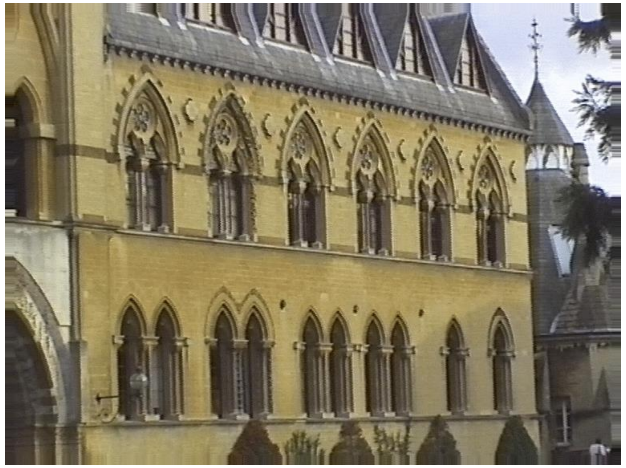
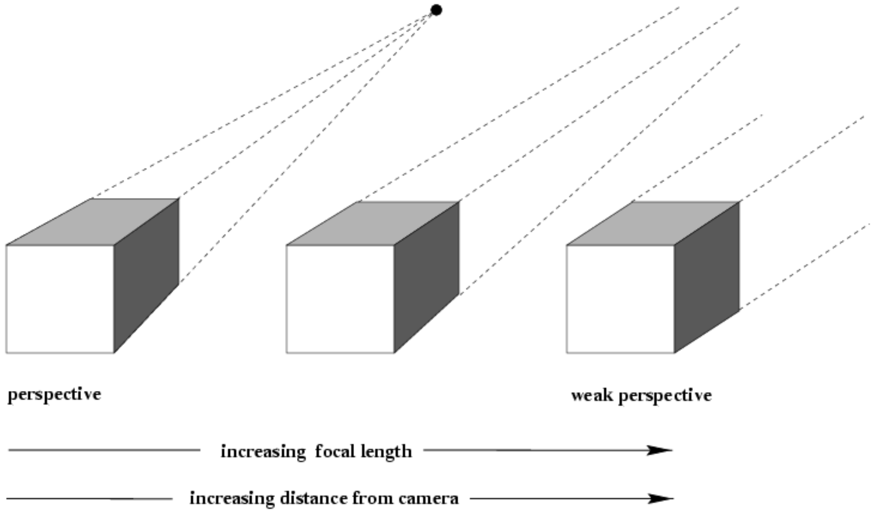


Figure source: [Wikipedia]

Orthographic or Parallel Projection

- What happens if we walk infinitely far away and zoom infinitely far in?

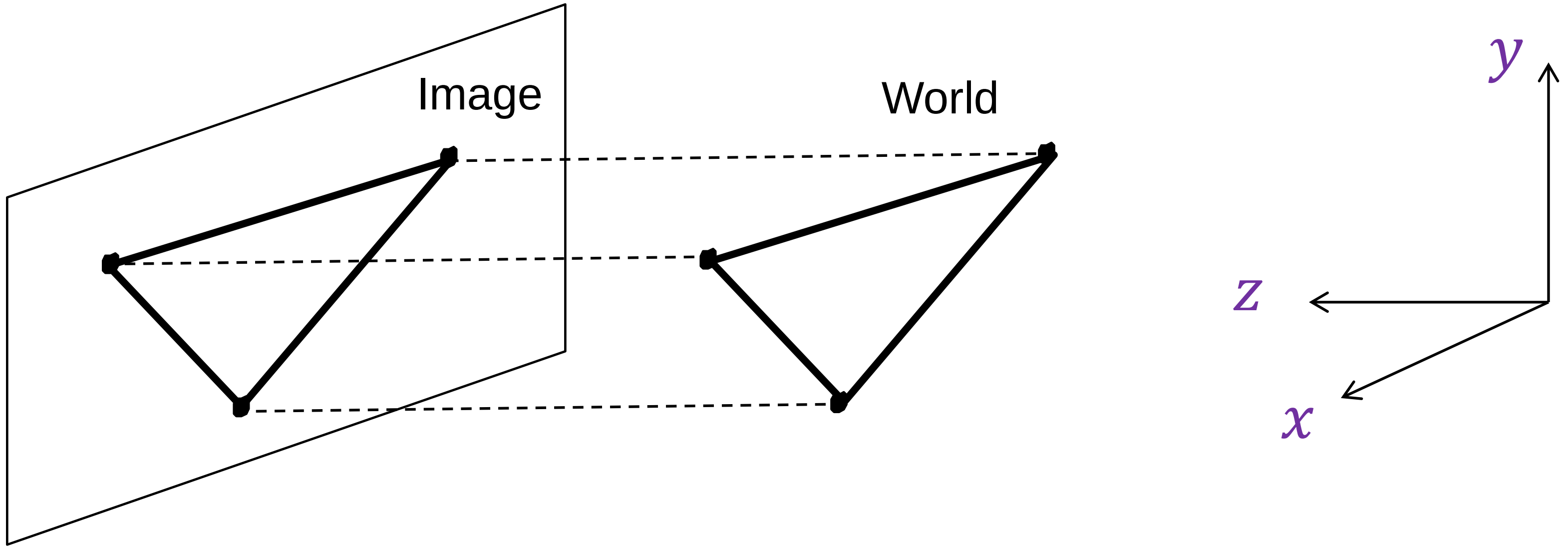




Source: Hartley & Zisserman

Orthographic or Parallel Projection

- Special case of perspective projection
  - Distance from center of projection to image plane is infinite
  - Projection equation: simply drop the z coordinate!



Slide by Steve Seitz

Orthographic or Parallel Projection

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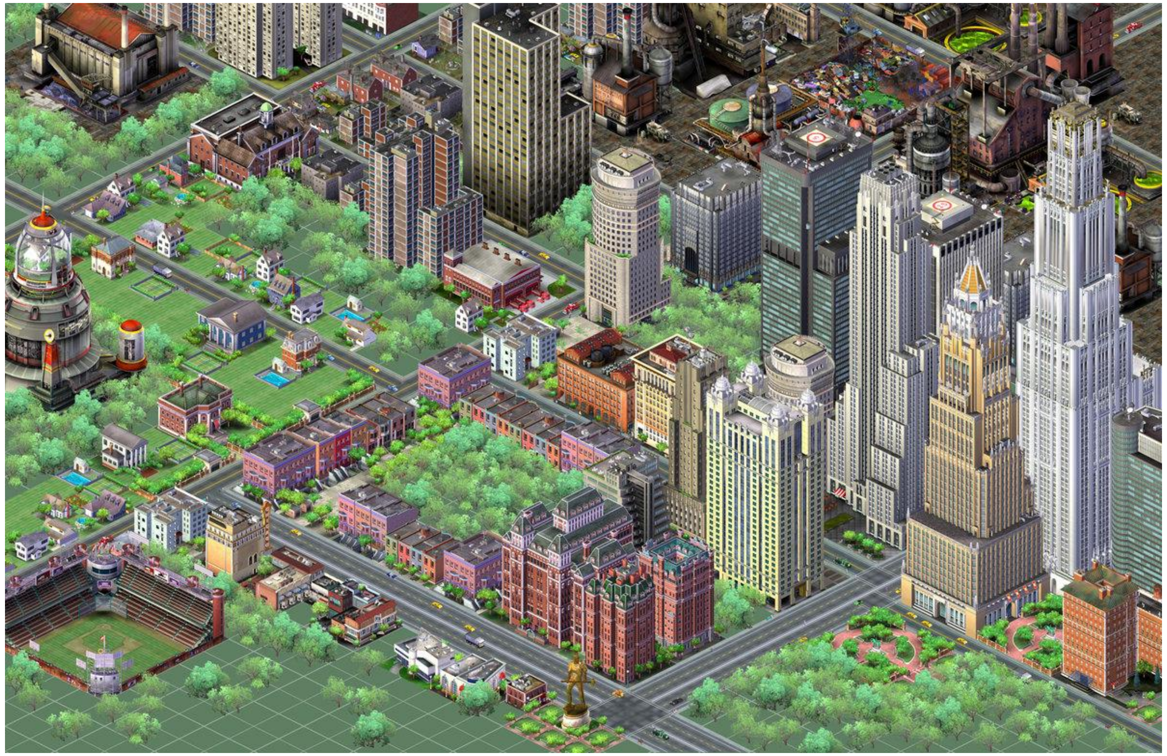
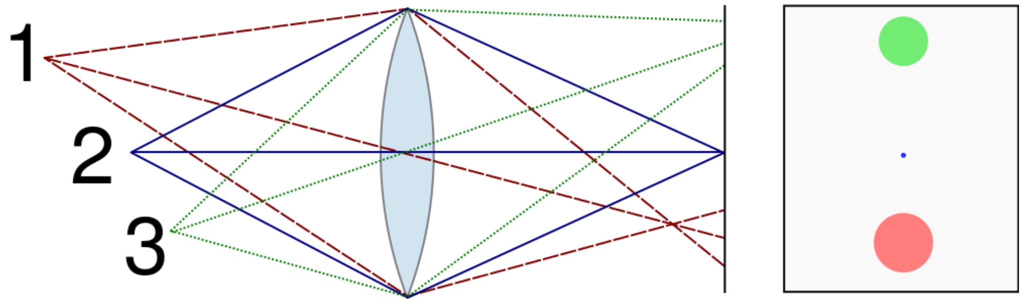


SimCity 2000

12th century Chinese scroll
(via [A. Hertzmann])

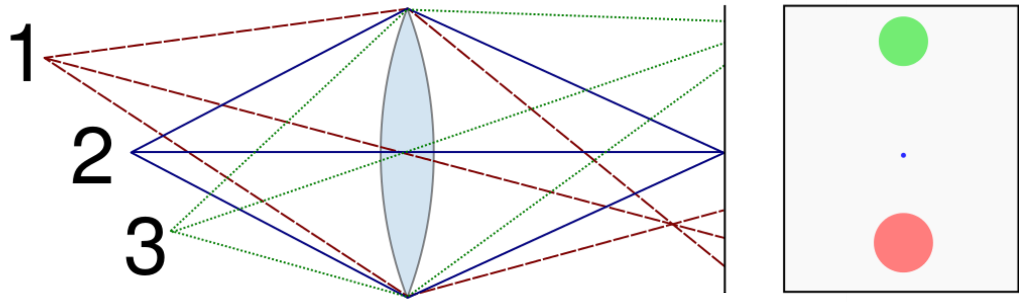
Cameras with Lenses

Cameras with Lenses



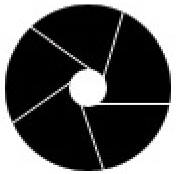
Cameras with Lenses

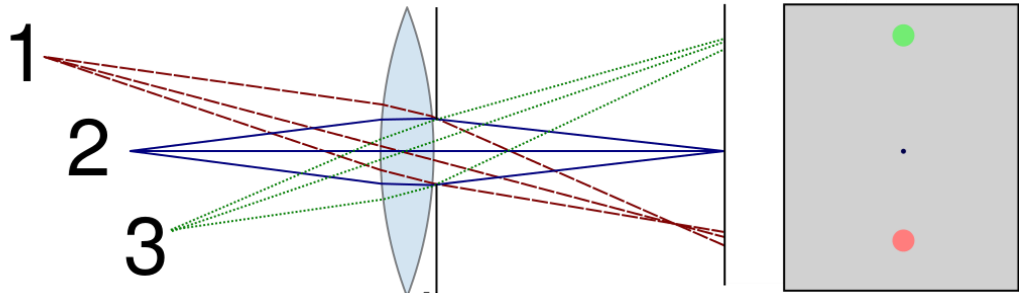
Large aperture



Small DOF

Small aperture



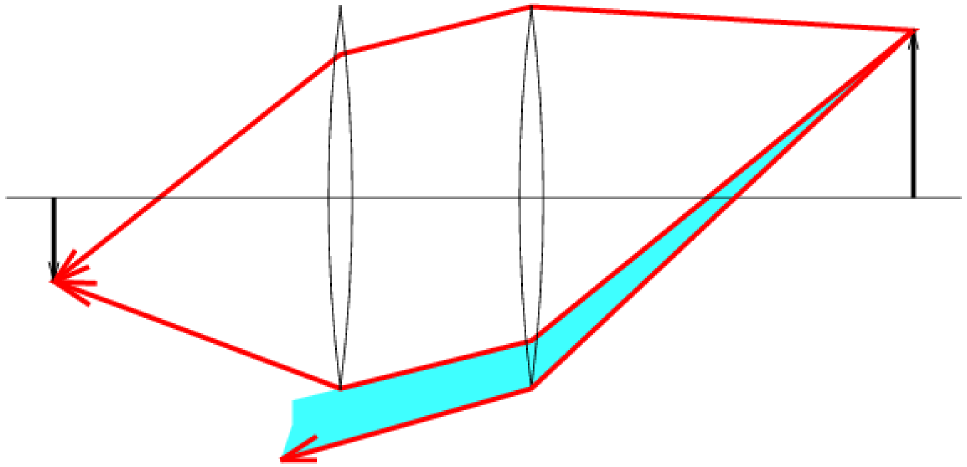


Large DOF

Lens Distortions

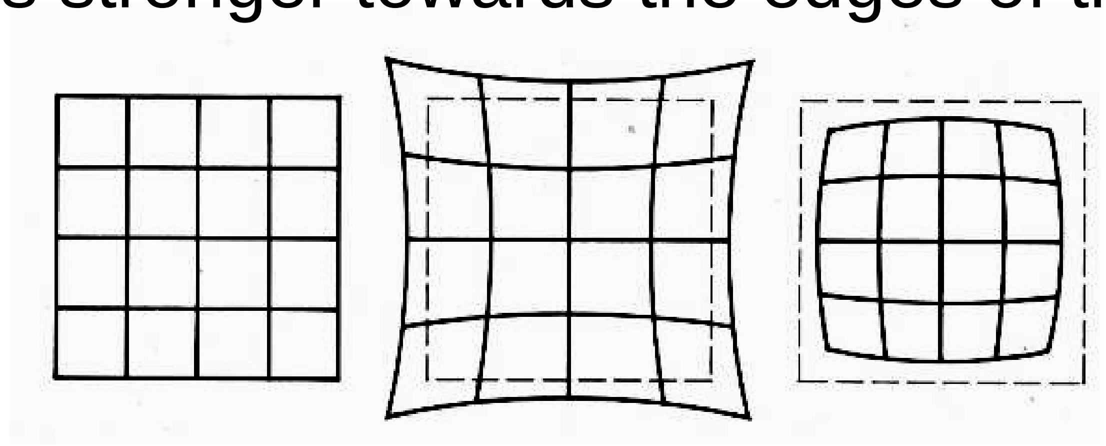
Lens Distortions

Vignetting



Radial Lens Distortions

- Caused by imperfect lenses
- Distortion is stronger towards the edges of the photo



No distortion Pin cushion Barrel

Radial Lens Distortions

Barrel

Pincushion

Fisheye

Modeling Distortions

Project $(\widehat{x}, \widehat{y}, \widehat{z})$ to "normalized" image coordinates

$$x'_n = \widehat{x}/\widehat{z}$$
$$y'_n = \widehat{y}/\widehat{z}$$

Apply radial distortion

$$r^2 = {x'_n}^2 + {y'_n}^2$$
$$x'_d = x'_n(1 + \kappa_1 r^2 + \kappa_2 r^4)$$
$$y'_d = y'_n(1 + \kappa_1 r^2 + \kappa_2 r^4)$$

Apply focal length translate image center

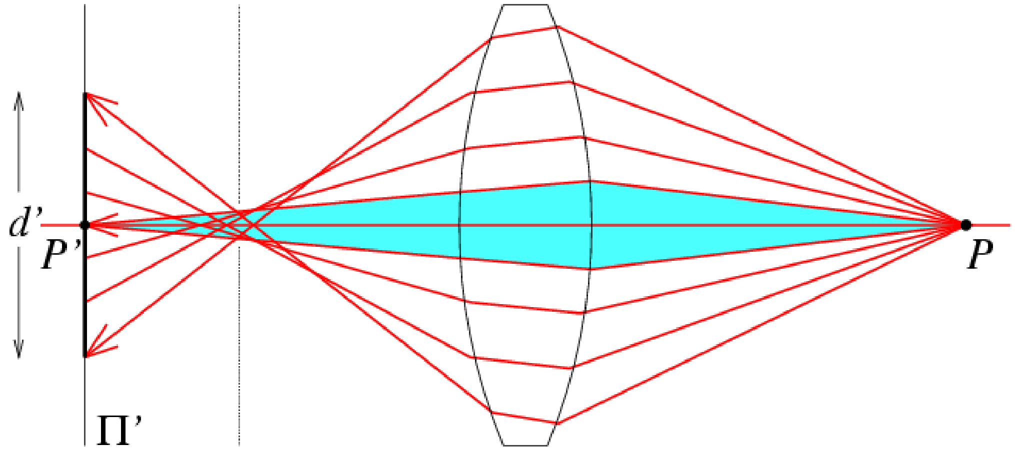
$$x' = f x'_d + x_c$$
$$y' = f y'_d + y_c$$

- To model lens distortion
  - Use above projection operation instead of standard projection matrix multiplication

Correcting Radial Distortions

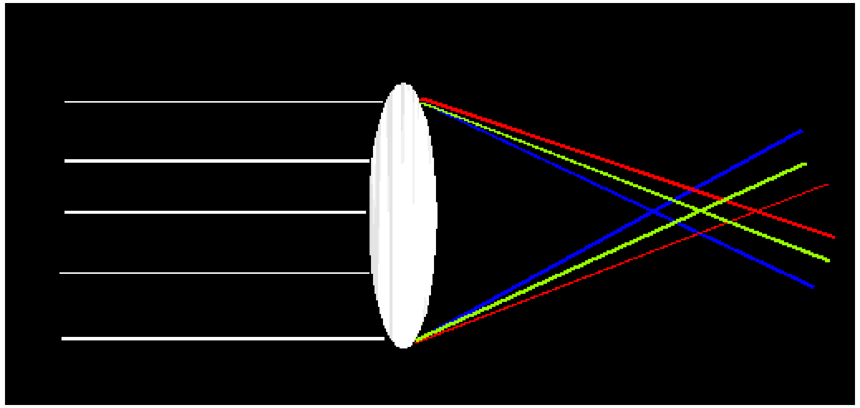
Spherical Aberration

- Spherical lenses don’t focus light perfectly
- Rays farther from the optical axis focus closer



Lens Flaws: Chromatic Aberration

- Lens has different refractive indices for different wavelengths: causes color fringing



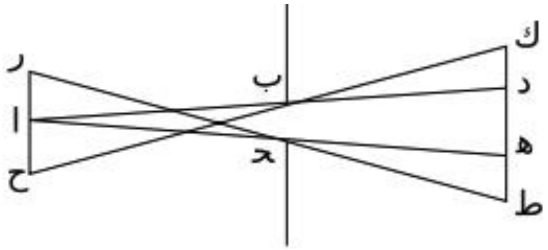
Near Lens Center

Near Lens Outer Edge

Historic Milestones

- **Pinhole model:** Mozi (470-390 BCE), Aristotle (384-322 BCE)
- **Principles of optics (including lenses):** Alhacen (965-1039 CE)
- **Camera obscura:** Leonardo da Vinci (1452-1519), Johann Zahn (1631-1707)
- **First photo:** Joseph Nicephore Niepce (1822)
- **Daguerréotypes** (1839)
- **Photographic film** (Eastman, 1889)
- **Cinema** (Lumière Brothers, 1895)
- **Color Photography** (Lumière Brothers, 1908)
- **Television** (Baird, Farnsworth, Zworykin, 1920s)
- **First consumer camera with CCD** Sony Mavica (1981)
- **First fully digital camera:** Kodak DCS100 (1990)

https://en.wikipedia.org/wiki/History_of_photography



Alhacen’s notes


Niepce, “La Table Servie,” 1822


Old television camera

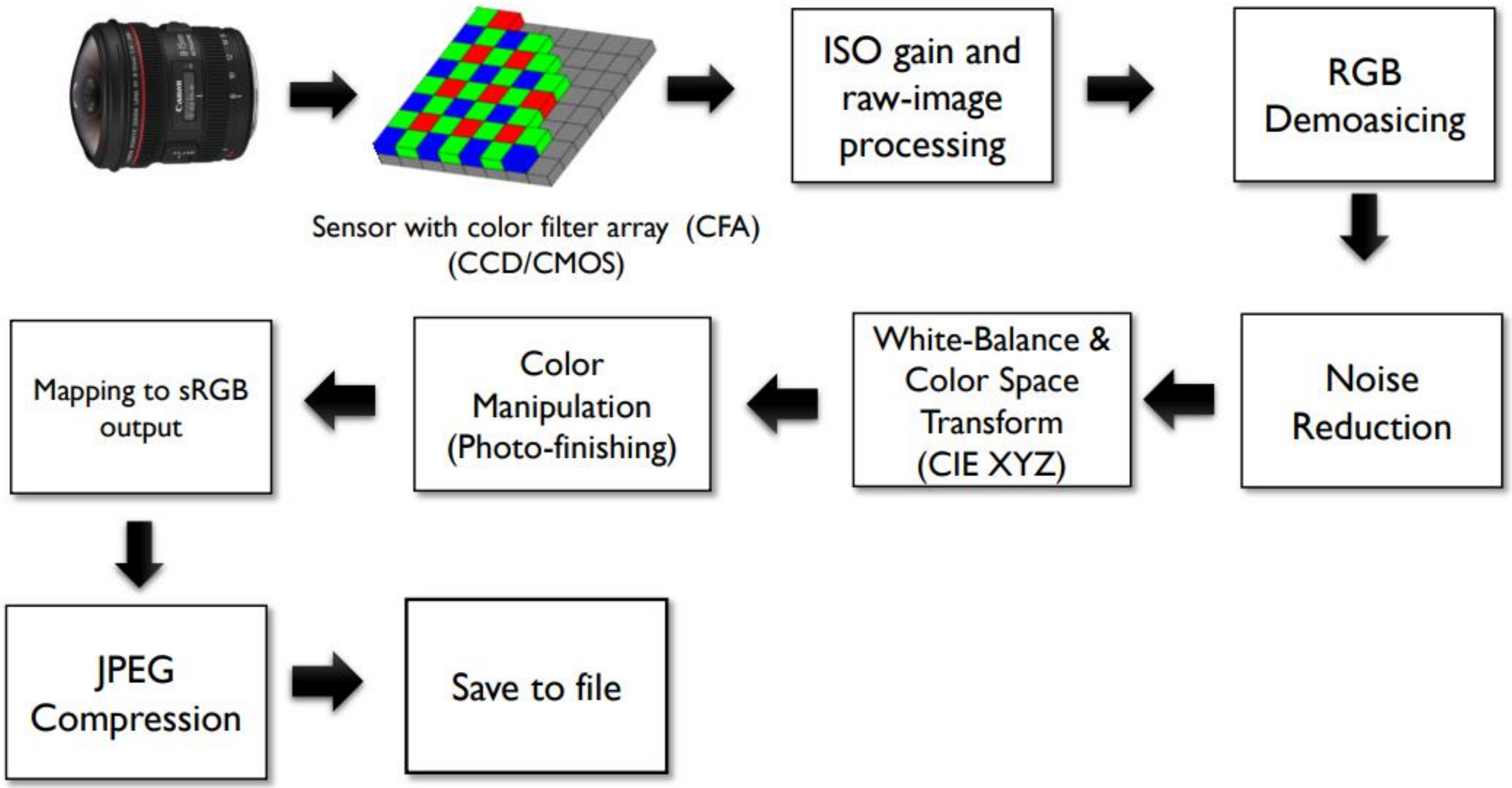
First Digitally Scanned Photograph

- NIST (1957), 176x176 pixels

http://listverse.com/history/top-10-incredible-early-firsts-in-photography/

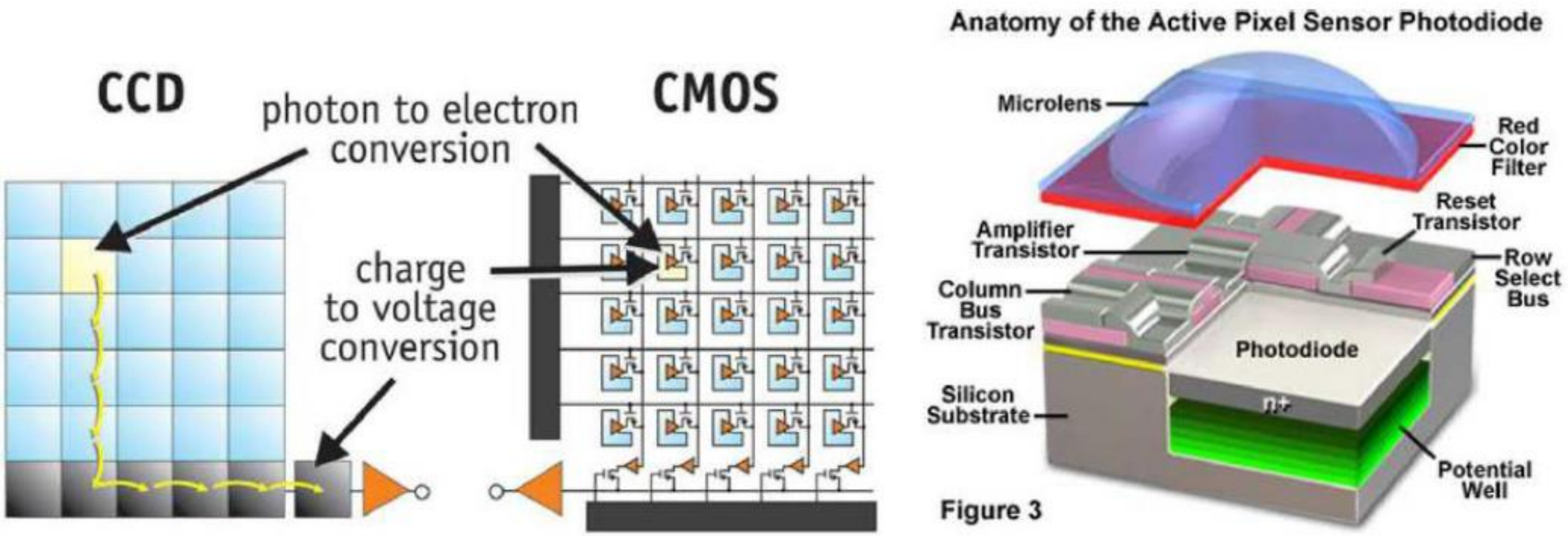
Digital Camera and Digital Sensors

Image Sensing Pipeline



NOTE:This diagram represents the steps applied on a typical consumer camera pipeline. ISPs may apply these steps in a different order or combine them in various ways. A modern camera ISP will undoubtedly be more complex, but will almost certainly implement these steps in some manner.

Digital Camera Sensors



(a) (b)

- Each cell in a sensor array is a light-sensitive diode that converts photons to electrons
  - Dominant in the past: **Charge Coupled Device (CCD)**
  - Dominant now: **Complementary Metal Oxide Semiconductor (CMOS)**

http://electronics360.globalspec.com/article/9464/ccd-vs-cmos-the-shift-in-image-sensor-technology

Camera Sensor

CMOS sensor

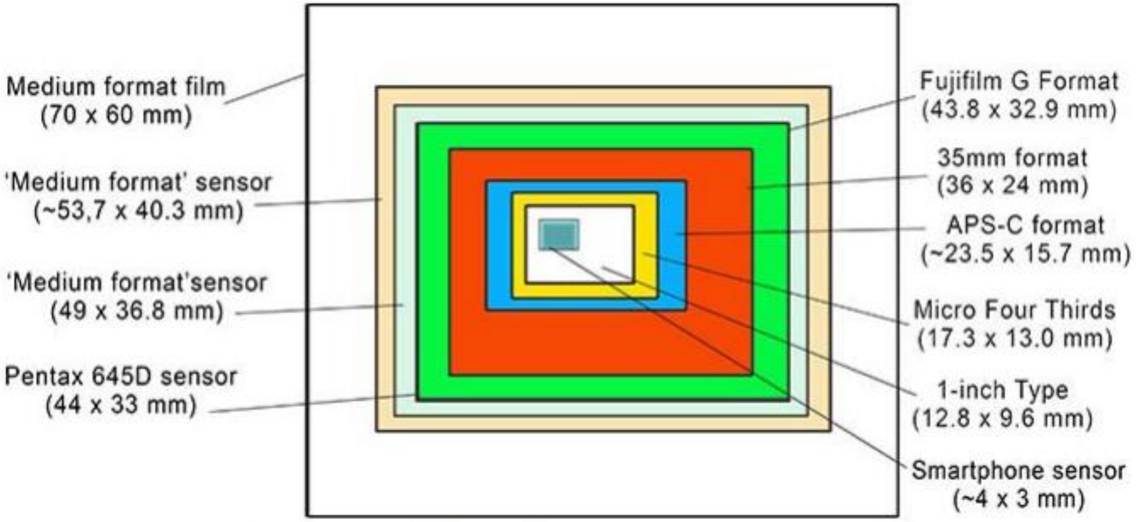


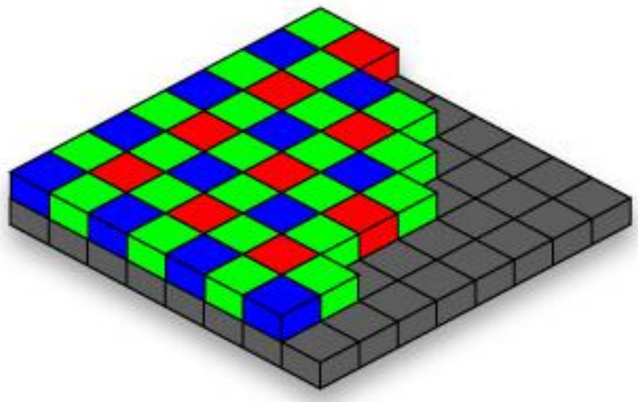
Figure from Photo Review website.

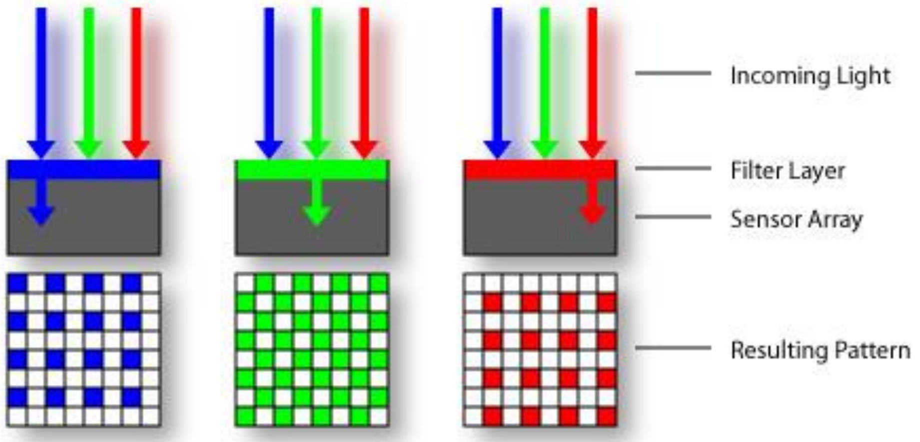
Almost all consumer camera sensors are based on complementary metal-oxide-semiconductor (CMOS) technology.

We generally describe sensors in terms of number of pixels and size. The larger the sensor, the better the noise performance as more light can fall on each pixel. Smart phones have small sensors!

Color Filter Arrays

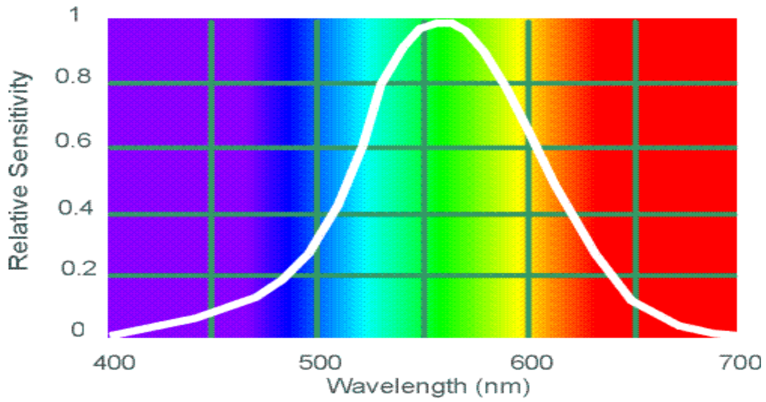
Bayer grid (1976)





Demosaicing:
Estimation of missing components from neighboring values

Why more green?

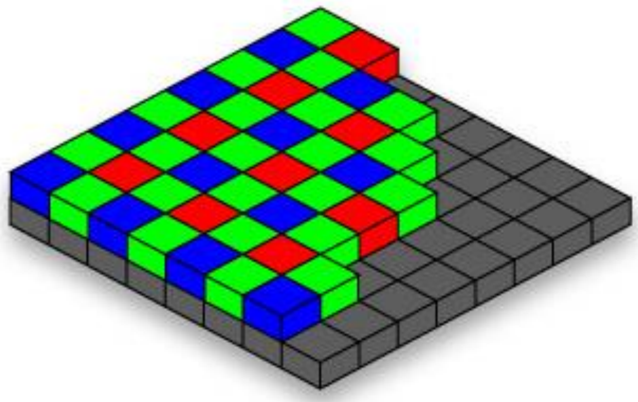


Human Luminance Sensitivity Function

Source: Steve Seitz

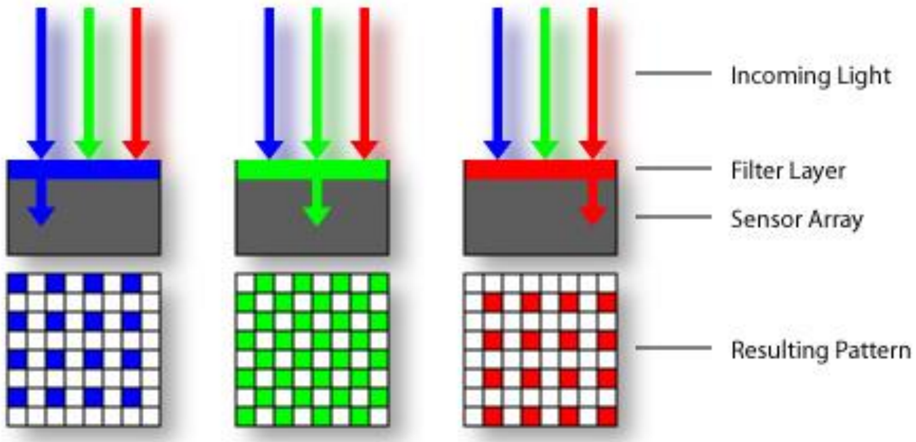
Color Filter Arrays

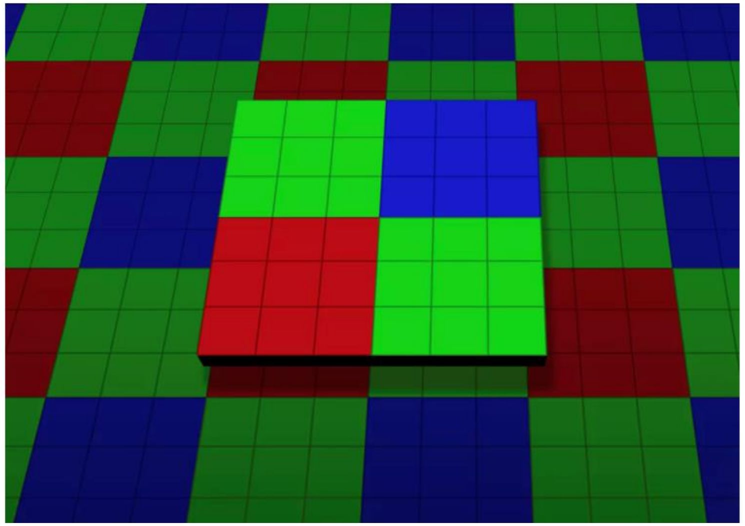
Bayer grid (1976)



Demosaicing:

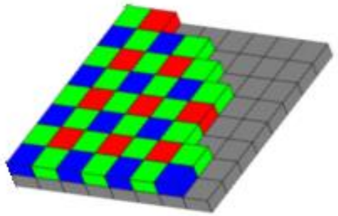
Estimation of missing components from neighboring values



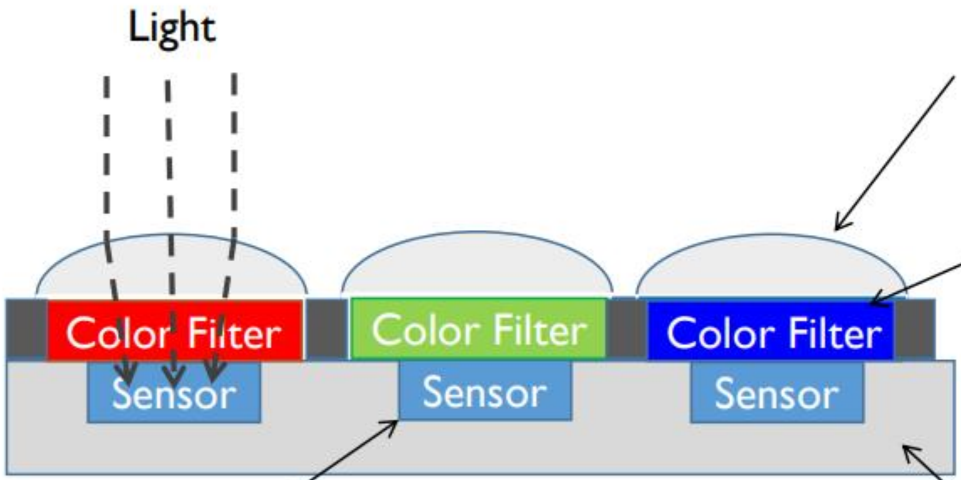


Recent cameraphone technology: pixel binning

Camera Sensor RGB Values



Color filter array
or "Bayer" pattern.



Micro-lenses are placed over the diode to help increase light collection on the sensor

Color filters place over the sensor. This forms a Color Filter Array (CFA) also called a "Bayer Pattern" after inventor Bryce Bayer.

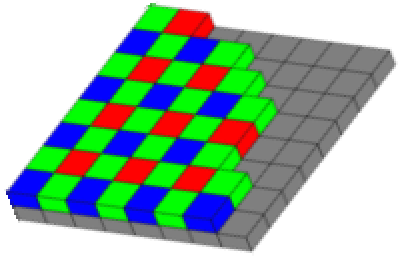
Photodiode
Photons hit the diode and force out electrons. This design is similar to a solar cell!

Silicon/Circuitry

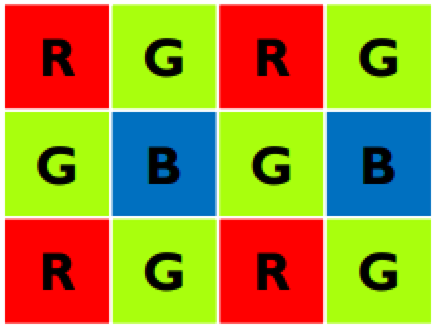
Bryce Bayer

CFA/Bayer Pattern Demosaicing

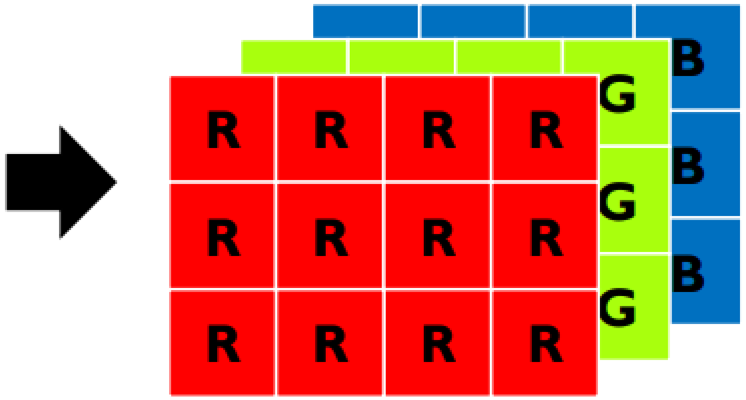
- Color filter array/Bayer pattern placed over pixel sensors
- We want an RGB value at each pixel, so we need to perform interpolation



Sensor with color filter array (CMOS)



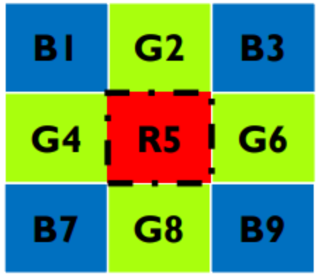
Sensor RGB layout



Desired output with RGB per pixel.

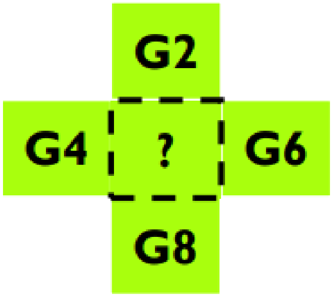
Simple Interpolation

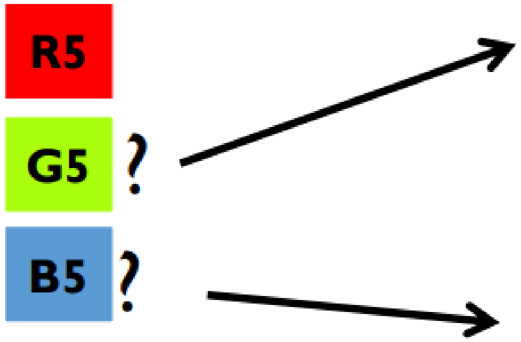
This is a zoomed up version of the Bayer pattern.



At location R5, we have a red pixel value, but no Green or Blue pixel.

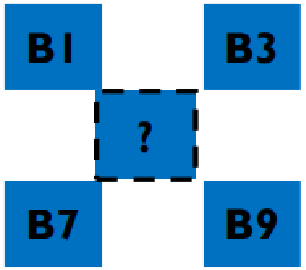
We need to estimate the G5 & B5 values at location R5.





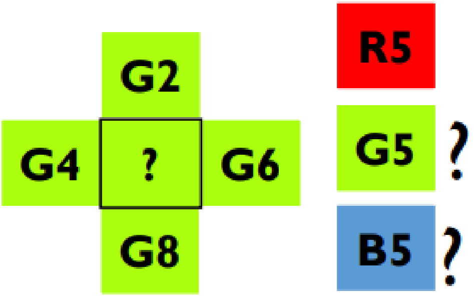
$$G5 = \frac{G2 + G4 + G6 + G8}{4}$$

$$B5 = \frac{B1 + B3 + B7 + B9}{4}$$



Camera Sensor RGB Values

B1	G2	B3
G4	R5	G6
B7	G8	B9



If (|G2–G8) && |(G4-G8)| both < Thres):

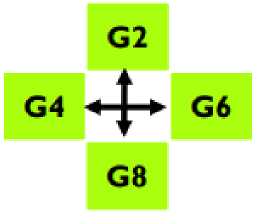
$$G5 = \frac{G2 + G4 + G6 + G8}{4}$$

elseif (|G2-G8| > Thres):

$$G5 = \frac{G4 + G6}{2}$$

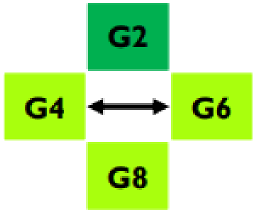
else:

$$G5 = \frac{G2 + G8}{2}$$



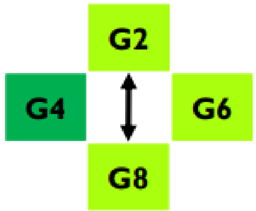
Case 1

All about the same.



Case 2

G2 and G8 differ – ignore them in the interpolation



Case 3

G2 and G8 differ – ignore them in the interpolation

Do this procedure also for the blue pixel B5

RGB Images (three channel)

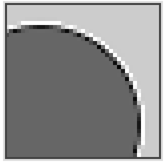
what we see

What we get out of the camera

Digital Camera Artifacts

Noise
- low light is where you most notice noise
- light sensitivity (ISO) / noise tradeoff
- stuck pixels

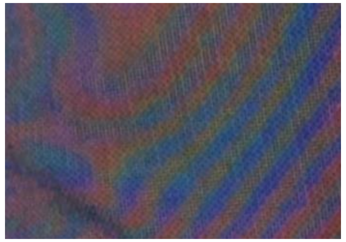
In-camera processing
- oversharpening can produce halos



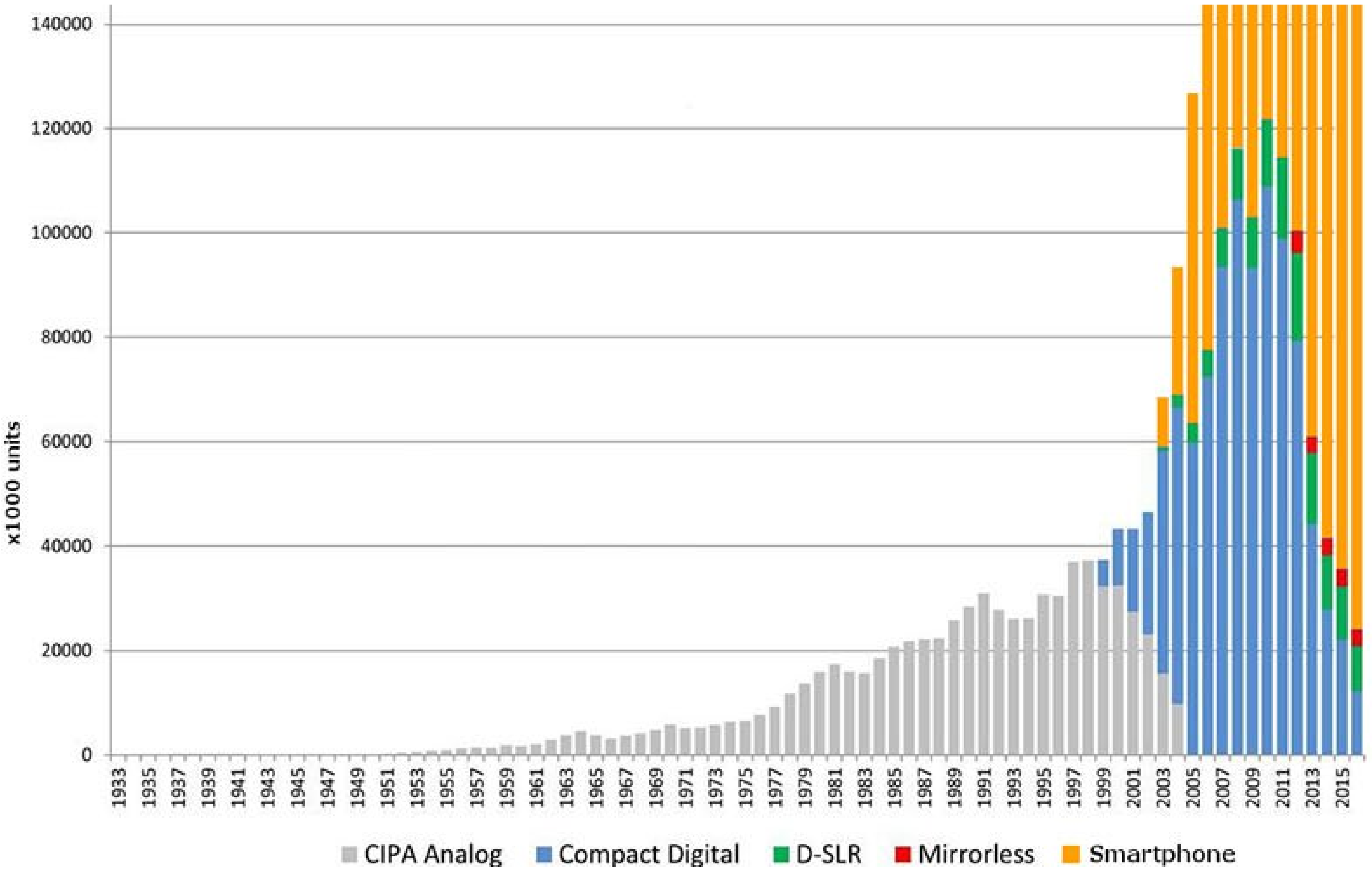
Compression
- JPEG artifacts, blocking

Blooming
- CCD charge overflowing into neighboring pixels

Color artifacts
- Color moire
- Purple fringing from microlenses



Camera Sales Over Time



[Source]

Camera Sales Over Time

The full chart…

Source

Thank You

Appendix

Digital Camera Pioneers

Bryce E. Bayer
(Color Filter Arrays)

Willard Boyle and George Smith
(Nobel Prize for inventing the CCD)
Photo: Reuters

Eric R. Fossum
(Invented CMOS)

Steven Sasson
(Attributed with building the first digital camera)